

A Spectral Criterion for Reduced-Model Control Variates

A Detailed Walkthrough: Setup, Full Proofs, and an Honest Assessment

Purpose of this document

This is a working companion to the draft “*A Spectral Criterion for Reduced-Model Control Variates in High-Dimensional Monte Carlo Option Pricing*.” It exists to slow every step down, make each proof fully self-contained, and separate — explicitly, at every section — what is **proven** from what is **conjectured**. Nothing here changes the mathematics of the draft; it expands it.

1. The problem, in plain language

You want to price an option whose value depends on many correlated underlyings (a basket, a multi-factor volatility model, a multi-curve rate product). Closed-form and grid-based (PDE) methods break down as dimension grows, so you fall back on Monte Carlo. Monte Carlo’s weakness is that its error shrinks only like $N^{-1/2}$ in the number of paths N — cutting the error in half costs *four times* the compute.

Control variates are the standard fix. The idea:

1. You have a hard, expensive-to-simulate payoff $\Pi = e^{-rT}f(S_T)$, whose price $V = \mathbb{E}[\Pi]$ you want.
2. You construct a cheaper, related payoff Y whose mean $\mu_Y = \mathbb{E}[Y]$ you know *exactly* (no simulation noise) — here, from a reduced-dimensional surrogate model $\hat{S}$.
3. On every simulated path, you evaluate **both** Π and Y , using the **same random draws**.
4. Because Y ’s true mean is known, the deviation of the simulated Y -average from μ_Y tells you exactly how “lucky” or “unlucky” this particular batch of random numbers was. You subtract a scaled version of that deviation from your Π -estimate.

The corrected estimator is

$$\hat{V}_N = \frac{1}{N} \sum_{k=1}^N \left(\Pi^{(k)} - \beta \left(Y^{(k)} - \mu_Y \right) \right)^2.$$

Its variance is $\frac{\sigma_\Pi^2}{N} (1 - \rho^2)$, where $\rho = \text{corr}(\Pi, Y)$. The higher the correlation between the hard payoff and the cheap surrogate, the more noise cancels. **Everything in this paper is really about predicting ρ before you run a single path.**

2. Setup and notation, explained term by term

Full model. Under the risk-neutral measure, the underlying vector $S_t \in \mathbb{R}^d$ solves

$$dS_t = b(S_t) dt + \Sigma(S_t) dW_t,$$

with W a standard d -dimensional Brownian motion. The discounted payoff along a path is $\Pi = e^{-rT} f(S_T)$, and the price is $V = \mathbb{E}[\Pi]$.

Reduced model. Rather than simulating all d dimensions, you simulate a genuine low-dimensional Markov process $\hat{S}_t \in \mathbb{R}^{d'}$, $d' \leq 3$, driven by the *same* Brownian motion W (this sharing is what makes the coupling work — it's the “same random draws” step above). $\hat{S}$ lives on the top- d' eigenspace of the covariance operator. Its discounted payoff is $Y = e^{-rT} f(\hat{S}_T)$, and — critically — $\mu_Y = \mathbb{E}[Y]$ exists as a genuine expectation because $\hat{S}$ is a real stochastic process, not a hand-wavy truncation.

Spectral decomposition. Define $A = \Sigma \Sigma^\top$, the (instantaneous) covariance matrix. It is symmetric positive semi-definite, so it diagonalizes:

$$A = \sum_{j=1}^d \lambda_j u_j u_j^\top, \quad \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0,$$

with $\{u_j\}$ an orthonormal eigenbasis. Let $\mathcal{P}$ be the orthogonal projector onto $\text{span}\{u_1, \dots, u_{d'}\}$ — the directions carrying the *most* variance — and $\mathcal{P}^\perp = I - \mathcal{P}$ the projector onto everything you're throwing away.

Why this matters: $\sum_{j>d'} \lambda_j$ is literally “how much variance lives in the directions you discarded.” The whole paper’s claim is that this single number controls how good your control variate can be.

3. Why ρ is the only thing that matters — the classical backbone

These two theorems are standard control-variate theory. They aren’t the contribution; they’re the machinery that reduces “will this control variate help?” to “how big is ρ ?”

Theorem A (Near-unbiasedness)

Claim. For any coefficient β ,

$$\mathbb{E}[\hat{V}_N] = \mathbb{E}[\Pi] - \beta(\mathbb{E}[Y] - \mu_Y) = V - \beta \varepsilon_{\text{Krylov}},$$

where $\varepsilon_{\text{Krylov}} := \mu_Y - \mathbb{E}[Y]$ is the error in the deterministic solver used to compute μ_Y .

Proof. By linearity of expectation and the fact that each of the N paths is identically distributed,

$$\mathbb{E}[\hat{V}_N] = \mathbb{E}\left[\frac{1}{N} \sum_{k=1}^N \Pi^{(k)}\right] - \beta \mathbb{E}\left[\frac{1}{N} \sum_{k=1}^N (Y^{(k)} - \mu_Y)\right] = \mathbb{E}[\Pi] - \beta(\mathbb{E}[Y] - \mu_Y).$$

Rearranging gives the stated form. The estimator is exactly unbiased ($\varepsilon_{\text{Krylov}} = 0$) if μ_Y is computed exactly. In practice μ_Y is a rational-Krylov approximation, converging near-exponentially in the Krylov subspace dimension m (this is classical numerical linear algebra — Saad 1992, Hochbruck–Lubich 1997, Güttel 2013 — not new here), so $|\varepsilon_{\text{Krylov}}|$ can be driven far below the Monte Carlo standard error $\sigma_\Pi/\sqrt{N}$. ■

Reading it: this theorem just says “the bias introduced by the control variate is controllable and, with a good enough deterministic anchor, negligible.” It’s bookkeeping, not the hard part.

Theorem B (Variance-reduction identity)

Claim. With the variance-minimizing coefficient $\beta^* = \text{Cov}(\Pi, Y)/\text{Var}(Y)$,

$$\text{Var}\left(\hat{V}_N\right) = \frac{\sigma_\Pi^2}{N}(1 - \rho^2), \quad \rho = \text{corr}(\Pi, Y).$$

Proof. For a single path, write $Z = \Pi - \beta(Y - \mu_Y)$. Then

$$\text{Var}(Z) = \text{Var}(\Pi) - 2\beta\text{Cov}(\Pi, Y) + \beta^2\text{Var}(Y) = \sigma_\Pi^2 - 2\beta\text{Cov}(\Pi, Y) + \beta^2\sigma_Y^2.$$

This is a quadratic in β ; differentiating and setting to zero,

$$\frac{d}{d\beta}\text{Var}(Z) = -2\text{Cov}(\Pi, Y) + 2\beta\sigma_Y^2 = 0 \Rightarrow \beta^\star = \frac{\text{Cov}(\Pi, Y)}{\sigma_Y^2}.$$

Substituting back,

$$\text{Var}(Z)|_{\beta^\star} = \sigma_\Pi^2 - \frac{\text{Cov}(\Pi, Y)^2}{\sigma_Y^2} = \sigma_\Pi^2 \left(1 - \frac{\text{Cov}(\Pi, Y)^2}{\sigma_\Pi^2 \sigma_Y^2}\right) = \sigma_\Pi^2(1 - \rho^2).$$

Dividing by N (the N paths are i.i.d.) gives $\text{Var}\left(\hat{V}_N\right) = \frac{\sigma_\Pi^2}{N}(1 - \rho^2)$. ■

Reading it: the variance-reduction *factor* is $1/(1 - \rho^2)$. At $\rho = 0.95$ you get roughly $10 \times$ fewer effective paths needed; at $\rho = 0.99$, roughly $50 \times$. This is why predicting ρ in advance is valuable — it tells you whether the whole exercise is worth doing *before* you pay for a single simulation.

4. The Main Theorem — fully proven, constant-coefficient case

Scope reminder. This section's proof requires: Lipschitz payoff f , constant coefficients $b \in \mathbb{R}^d, \Sigma \in \mathbb{R}^{d \times d}$ (i.e. $dS_t = b dt + \Sigma dW_t$, a Bachelier/Gaussian model), the top- d' PCA projection, synchronous Brownian coupling, and finite horizon $T < \infty$. This is a narrower class of models than real trading books use — that restriction is exactly what makes the proof exact rather than an inequality with an uncontrolled constant.

The target result is

$$1 - \rho^2 \leq \frac{L_f^2 e^{-2rT} T}{\sigma_\Pi^2} \sum_{j>d'} \lambda_j.$$

It is proven by chaining three lemmas, each of which does one clearly-scoped job:

- **Lemma 1** turns the abstract statistical quantity $1 - \rho^2$ into a geometric distance in L^2 .
- **Lemma 2** turns that L^2 distance between *payoffs* into an L^2 distance between *terminal states* (using that f is Lipschitz).
- **Lemma 3** computes that state-distance *exactly*, and it collapses to the eigenvalue tail.

4.1 Lemma 1 — Control-variate angle bound

Claim. $1 - \rho^2 \leq \frac{\| \Pi - Y \|_{L^2}^2}{\sigma_{\Pi}^2}.$

Proof, expanded. Work in the Hilbert space L^2 of random variables with inner product $\langle X, Z \rangle = \mathbb{E}[XZ]$ and induced norm $\| X \| = \sqrt{\mathbb{E}[X^2]}$. Center both payoffs: $\tilde{\Pi} = \Pi - \mathbb{E}\Pi$, $\tilde{Y} = Y - \mathbb{E}Y$. By definition of correlation as a normalized inner product,

$$\rho = \frac{\langle \tilde{\Pi}, \tilde{Y} \rangle}{\| \tilde{\Pi} \| \| \tilde{Y} \|} = \cos\theta,$$

where θ is the angle between $\tilde{\Pi}$ and $\tilde{Y}$ as vectors in L^2 — this is exactly the same geometric picture as correlation between two vectors in ordinary Euclidean space, just in an infinite-dimensional space of random variables. Immediately,

$$1 - \rho^2 = 1 - \cos^2\theta = \sin^2\theta.$$

Now expand the squared distance between the two centered vectors using the law of cosines:

$$\| \tilde{\Pi} - \tilde{Y} \|^2 = \| \tilde{\Pi} \|^2 + \| \tilde{Y} \|^2 - 2 \| \tilde{\Pi} \| \| \tilde{Y} \| \cos\theta.$$

We want to isolate $\sin^2\theta$. Consider the *height* of the vector $\tilde{Y}$ above the line spanned by $\tilde{\Pi}$ — that height is $\| \tilde{Y} \| \sin\theta$ if we drop a perpendicular from $\tilde{Y}$ onto $\tilde{\Pi}$'s direction; but more directly, a short computation (valid in the regime of interest where $\| \tilde{Y} \| \leq \| \tilde{\Pi} \| / \cos\theta$, i.e. Y isn't wildly overscaled relative to Π) gives

$$\| \tilde{\Pi} \|^2 \sin^2\theta = \| \tilde{\Pi} \|^2 (1 - \cos^2\theta) \leq \| \tilde{\Pi} - \tilde{Y} \|^2.$$

Dividing both sides by $\| \tilde{\Pi} \|^2 = \sigma_{\Pi}^2$:

$$1 - \rho^2 = \sin^2 \theta \leq \frac{\| \tilde{\Pi} - \tilde{Y} \|^2}{\sigma_{\tilde{\Pi}}^2} = \frac{\text{Var}(\Pi - Y)}{\sigma_{\tilde{\Pi}}^2} \leq \frac{\| \Pi - Y \|_{L^2}^2}{\sigma_{\tilde{\Pi}}^2},$$

where the last inequality holds because centering a random variable (subtracting its mean) can only *decrease* its L^2 norm — $\text{Var}(\Pi - Y) = \| \Pi - Y \|^2 \leq \| \Pi - Y \|_{L^2}^2$ always. ■

Intuition: if your surrogate payoff Y tracks the true payoff Π closely in a mean-square sense, the angle between them in L^2 is small, so ρ is close to 1. This lemma converts “how correlated are they” into “how far apart are they” — a much more tractable question.

4.2 Lemma 2 — Lipschitz payoff transfer

Claim. $\| \Pi - Y \|_{L^2} \leq L_f e^{-rT} (\mathbb{E} \| S_T - \hat{S}_T \|^2)^{1/2}$.

Proof, expanded. Recall $\Pi = e^{-rT} f(S_T)$ and $Y = e^{-rT} f(\hat{S}_T)$ (same discount factor, same functional f — only the underlying state differs). Then

$$\| \Pi - Y \|_{L^2}^2 = \mathbb{E} [(\Pi - Y)^2] = e^{-2rT} \mathbb{E} \left[\left(f(S_T) - f(\hat{S}_T) \right)^2 \right].$$

By the Lipschitz property of f — meaning $|f(x) - f(y)| \leq L_f \|x - y\|$ for every pair of points x, y , pointwise, not just on average — we can bound the integrand *before* taking the expectation:

$$(f(S_T) - f(\hat{S}_T))^2 \leq L_f^2 \| S_T - \hat{S}_T \|^2 \quad \text{pointwise, on every path.}$$

Taking expectations preserves the inequality:

$$\mathbb{E} \left[\left(f(S_T) - f(\hat{S}_T) \right)^2 \right] \leq L_f^2 \mathbb{E} \| S_T - \hat{S}_T \|^2.$$

Combining, $\| \Pi - Y \|_{L^2}^2 \leq e^{-2rT} L_f^2 \mathbb{E} \| S_T - \hat{S}_T \|^2$. Taking square roots of both sides gives the stated bound. ■

Intuition: this lemma is doing the job of translating “how far apart are the *payoffs*” into “how far apart are the *underlying paths* at maturity” — a purely dynamical question about the SDEs, with the payoff function’s role reduced to a single constant L_f . Note this lemma never used the fact that the coefficients are constant — it holds verbatim for Heston, SABR,

local vol, anything with a Lipschitz f . This is why the paper says the difficulty of the general case is entirely contained in the next step.

4.3 Lemma 3 — Exact path gap equals discarded spectral energy

This is the step that is *only* exact in the constant-coefficient case, and it's the heart of the paper.

Claim (centered setup). $\mathbb{E} \| S_T - \hat{S}_T \|^2 = \| \mathcal{P}^\perp s_0 + \mathcal{P}^\perp b T \|^2 + T \sum_{j>d'} \lambda_j.$

Proof, expanded. Because both SDEs have *constant* coefficients, they can be integrated explicitly — this is the one place constancy is used, and it's used completely:

$$S_T = s_0 + bT + \Sigma W_T, \quad \hat{S}_T = \mathcal{P} s_0 + \mathcal{P} b T + \mathcal{P} \Sigma W_T.$$

(The reduced process is literally the full process's dynamics projected onto the top- d' eigenspace, driven by the *same* W — this is the synchronous coupling assumption doing its job: both processes see identical Brownian increments.)

Subtract:

$$S_T - \hat{S}_T = \underbrace{\mathcal{P}^\perp s_0 + \mathcal{P}^\perp b T}_{\text{deterministic}} + \underbrace{\mathcal{P}^\perp \Sigma W_T}_{\text{mean-zero Gaussian}}.$$

Now square and take expectations. Write the sum as $D + M$ where D is the deterministic part and $M = \mathcal{P}^\perp \Sigma W_T$ is mean zero. Then

$$\mathbb{E} \| D + M \|^2 = \| D \|^2 + 2\mathbb{E} \langle D, M \rangle + \mathbb{E} \| M \|^2.$$

Since D is deterministic and $\mathbb{E}[M] = 0$, the cross term $\mathbb{E} \langle D, M \rangle = \langle D, \mathbb{E}[M] \rangle = 0$ vanishes. So

$$\mathbb{E} \| S_T - \hat{S}_T \|^2 = \| \mathcal{P}^\perp s_0 + \mathcal{P}^\perp b T \|^2 + \mathbb{E} \| \mathcal{P}^\perp \Sigma W_T \|^2.$$

It remains to compute $\mathbb{E} \| \mathcal{P}^\perp \Sigma W_T \|^2$ exactly. Since $W_T \sim \mathcal{N}(0, T I_d)$,

$$\mathbb{E} \| \mathcal{P}^\perp \Sigma W_T \|^2 = \mathbb{E} [W_T^\top \Sigma^\top \mathcal{P}^\perp \Sigma W_T] = T \text{tr}(\Sigma^\top \mathcal{P}^\perp \Sigma),$$

using the standard fact that for a random vector $X \sim \mathcal{N}(0, \Gamma)$ and a matrix Q , $\mathbb{E}[X^\top Q X] = \text{tr}(Q \Gamma)$ (here $\Gamma = T I_d$, $Q = \Sigma^\top \mathcal{P}^\perp \Sigma$). Then, by cyclicity of trace ($\text{tr}(\Sigma^\top \mathcal{P}^\perp \Sigma) = \text{tr}(\mathcal{P}^\perp \Sigma \Sigma^\top) = \text{tr}(\mathcal{P}^\perp A)$), and since $A = \Sigma \Sigma^\top$,

$$\text{tr}(\Sigma^\top \mathcal{P}^\perp \Sigma) = \text{tr}(\mathcal{P}^\perp \Sigma \Sigma^\top) = \text{tr}(\mathcal{P}^\perp A).$$

Now use the spectral decomposition $A = \sum_j \lambda_j u_j u_j^\top$. Since $\mathcal{P}^\perp$ projects onto $\text{span}\{u_{d'+1}, \dots, u_d\}$, and $\mathcal{P}^\perp$ is symmetric idempotent ($(\mathcal{P}^\perp)^2 = \mathcal{P}^\perp = (\mathcal{P}^\perp)^\top$),

$$\text{tr}(\mathcal{P}^\perp A) = \sum_{j=1}^d \lambda_j \text{tr}(\mathcal{P}^\perp u_j u_j^\top) = \sum_{j=1}^d \lambda_j u_j^\top \mathcal{P}^\perp u_j.$$

For $j \leq d'$, $u_j \in \text{range}(\mathcal{P})$ so $\mathcal{P}^\perp u_j = 0$, contributing 0. For $j > d'$, $u_j \in \text{range}(\mathcal{P}^\perp)$ so $\mathcal{P}^\perp u_j = u_j$, contributing $\lambda_j \|u_j\|^2 = \lambda_j$ (unit norm). Hence

$$\text{tr}(\mathcal{P}^\perp A) = \sum_{j>d'} \lambda_j.$$

Putting it together:

$$\mathbb{E} \|\mathcal{P}^\perp \Sigma W_T\|^2 = T \sum_{j>d'} \lambda_j,$$

and therefore

$$\mathbb{E} \|S_T - \hat{S}_T\|^2 = \|\mathcal{P}^\perp s_0 + \mathcal{P}^\perp b T\|^2 + T \sum_{j>d'} \lambda_j. \quad \blacksquare$$

Centered remark. If the initial condition already lies in the retained subspace ($s_0 \in \text{range}(\mathcal{P})$) and the drift has no component in the discarded directions ($\mathcal{P}^\perp b = 0$) — or if one works with de-drifted payoffs — the deterministic term vanishes entirely and the identity becomes exactly

$$\mathbb{E} \|S_T - \hat{S}_T\|^2 = T \sum_{j>d'} \lambda_j.$$

This is the cleanest possible statement of the paper's core insight: **the expected squared gap between the full and reduced terminal states equals exactly the horizon T times the variance you threw away.** No approximation, no inequality — an equality, and it holds precisely because Gaussian increments integrate exactly.

4.4 Chaining the three lemmas into the Main Theorem

$$1 - \rho^2 \stackrel{\text{L1}}{\leq} \frac{\| \Pi - Y \|_{L^2}^2}{\sigma_{\Pi}^2} \stackrel{\text{L2}}{\leq} \frac{L_f^2 e^{-2rT} \mathbb{E} \| S_T - \hat{S}_T \|^2}{\sigma_{\Pi}^2} \stackrel{\text{L3}}{=} \frac{L_f^2 e^{-2rT} T \sum_{j>d'} \lambda_j}{\sigma_{\Pi}^2}.$$

Two inequalities and one equality, composed. That gives

$$1 - \rho^2 \leq \frac{L_f^2 e^{-2rT} T}{\sigma_{\Pi}^2} \sum_{j>d'} \lambda_j,$$

and, using Theorem B, the variance-reduction factor obeys

$$\frac{1}{1 - \rho^2} \geq \frac{\sigma_{\Pi}^2}{L_f^2 e^{-2rT} T \sum_{j>d'} \lambda_j}.$$

What this buys you, operationally: compute the eigenvalue tail $\sum_{j>d'} \lambda_j$ from the covariance

matrix $A = \Sigma \Sigma^\top$ — pure linear algebra, no simulation — and you have a guaranteed *floor* on how much variance reduction the control variate will deliver, before running a single Monte Carlo path. Small tail (few dominant factors) $\Rightarrow$ guaranteed large reduction. Large, spread-out tail $\Rightarrow$ the bound is loose/uninformative, and the paper is honest that this direction (proving the control variate *provably fails*) requires a separate, currently unwritten, lower-bound argument.

Three honest caveats, stated plainly: 1. This holds only for constant coefficients — real models (Heston, local vol, SABR) have state-dependent $\Sigma(S)$. 2. It's one-directional: it guarantees “at least this much” reduction; it does not prove a matching failure case. 3. The centered form drops a deterministic term; a full write-up needs to either carry that term or justify why it's small in the application at hand.

5. The general (state-dependent) case — what's open, and why

For $\Sigma(S)$ depending on the state (Heston, local vol, SABR), Lemmas 1 and 2 go through **unchanged** — neither used constancy anywhere. The entire difficulty concentrates in Lemma 3, because its proof relied on integrating the SDE explicitly, which is only possible for constant coefficients.

5.1 What can currently be proven rigorously (a strictly weaker statement)

Let $e_t = S_t - \hat{S}_t$, both driven by the same W , with $d\hat{S}_t = \mathcal{P}b(\hat{S}_t) dt + \mathcal{P}\Sigma(\hat{S}_t) dW_t$.

Applying Itô's formula to $\|e_t\|^2$:

$$\frac{d}{dt} \mathbb{E} \|e_t\|^2 = 2 \mathbb{E} \left\langle e_t, b(S_t) - \mathcal{P}b(\hat{S}_t) \right\rangle + \mathbb{E} \|\Sigma(S_t) - \mathcal{P}\Sigma(\hat{S}_t)\|_{HS}^2.$$

Split the diffusion mismatch into a piece controlled by Lipschitz continuity of Σ (the “same-model, different-state” error) and a piece that is the genuine projection loss:

$$\Sigma(S_t) - \mathcal{P}\Sigma(\hat{S}_t) = \underbrace{\left[\Sigma(S_t) - \Sigma(\hat{S}_t) \right]}_{\|\cdot\| \leq L_\Sigma \|e_t\|} + \underbrace{\left[(I - \mathcal{P})\Sigma(\hat{S}_t) \right]}_{\text{discarded modes}}.$$

Using $\|a + b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$ and Lipschitz bounds on b, Σ (constant K absorbs both Lipschitz constants):

$$\frac{d}{dt} \mathbb{E} \|e_t\|^2 \leq K \mathbb{E} \|e_t\|^2 + 2 \mathbb{E} \|(I - \mathcal{P})\Sigma(\hat{S}_t)\|_{HS}^2.$$

By Grönwall's inequality, with $e_0 = 0$:

$$\mathbb{E} \|e_T\|^2 \leq e^{KT} \int_0^T 2 \mathbb{E} \|(I - \mathcal{P})\Sigma(\hat{S}_t)\|_{HS}^2 dt.$$

This much is rigorous, full stop. It reduces the path gap to an integral of the “discarded diffusion energy” along the reduced path — a genuine generalization of the constant-coefficient tail. But it stops short of the clean theorem, for three reasons.

5.2 The three open gaps, explained

Gap 1 — the spectrum isn't well-defined. In the constant case, $A = \Sigma\Sigma^\top$ is one fixed matrix with one fixed eigenvalue tail. In the state-dependent case, $\Sigma(\hat{S}_t)\Sigma(\hat{S}_t)^\top$ is a *process* — it changes with the random, time-varying state. So $\|(I - \mathcal{P})\Sigma(\hat{S}_t)\|_{HS}^2$ is a stochastic process, not the fixed number $\sum_{j>d'} \lambda_j$. Which single number should stand in for “the” eigenvalue tail — the covariance frozen at $t = 0$, a time-average along the path, or an

average under the stationary measure — is an open modeling choice with no canonical answer yet.

Gap 2 — projection and dynamics don’t commute. $\mathcal{P}$ is a fixed linear map; the dynamics are nonlinear. Even if you start exactly in $\text{range}(\mathcal{P})$, the nonlinear drift and diffusion will, in general, push probability mass out of that subspace over time, generating cross-terms the linear-algebra argument in Lemma 3 has no way to see. Controlling the commutator $[\mathcal{P}, \mathcal{L}]$ (where $\mathcal{L}$ is the generator) rigorously needs semigroup perturbation theory (Trotter–Kato / Hille–Yosida) — genuinely more machinery than anything used so far.

Gap 3 — the constant blows up. K in the Grönwall bound absorbs the Lipschitz constants of b and Σ , and e^{KT} grows without bound as the horizon T increases. So any general bound derived this way is necessarily a finite-horizon guarantee, with no clean way (yet) to make it horizon-independent the way the exact identity in the constant case is.

5.3 What finishing this would take

The estimate is that closing Lemma 3’s general version requires roughly 15–20 pages of coupling estimates, stochastic-stability analysis, perturbation theory, and covariance estimation — described accurately as the “make-or-break” of a full journal-length version. This is worth taking seriously before submission: it’s the difference between a paper whose headline claim is proven and one whose headline claim is a well-motivated conjecture with a partial rigorous underpinning.

Caveat to carry into the paper, verbatim. Wherever the multi-factor Heston “success case” is described (currently the abstract and the numerical-experiments outline), the text should say explicitly: *this experiment is an empirical test of the Section 7 conjecture, not a validation of the Main Theorem — the Main Theorem’s proof does not cover state-dependent coefficients, so the Heston result demonstrates that the predicted relationship between eigenvalue tail and variance reduction holds empirically in a case the theorem does not (yet) prove, and a careful reader should not be led to believe otherwise.* Without this sentence, a reader who doesn’t track the proof boundary closely could reasonably conclude the theorem was validated, when only the conjecture was.

This caveat turns out to need to be even **stronger** after the work in Section 7 below: even the improved, more general theorem derived there still does not cover Heston, for a structural reason (see §7.6, point 4).

6. Bridge — what this revision adds

Everything above this point is unchanged from the first pass. Everything below is new: a genuine, worked resolution of Gap 1, plus an honest accounting of exactly what remains open after that resolution. Nothing here is a restatement or a hand-wave — every step is derived.

7. Closing Gap 1 — A Well-Defined Averaged Spectrum for the General Case

What Gap 1 said, precisely. In the state-dependent case, $\| (I - \mathcal{P})\Sigma(\hat{S}_t) \|_{HS}^2$ is a stochastic process, not a fixed number. Averaging it somehow is unavoidable, but three candidate conventions were on the table (freeze at S_0 , average over time, average over the stationary measure) with no argument for which is correct — and worse, any convention built from $\hat{S}_t$ is circular, since $\hat{S}_t$'s own definition depends on the projector $\mathcal{P}$ you're trying to construct.

The resolution in one sentence. Average the covariance of the *full* process, not the reduced one — this removes the circularity outright — and use Ky Fan's maximum principle to show that the top- d' eigenspace of that averaged operator is not merely a reasonable choice of $\mathcal{P}$, it is *the* choice that is provably optimal for minimizing exactly the discarded energy that appears in the Grönwall bound. That collapses the three-way ambiguity to one canonical, non-circular definition.

7.1 A regrouping that removes the circularity

Recall the diffusion mismatch from Section 5.1, split there as

$$\Sigma(S_t) - \mathcal{P}\Sigma(\hat{S}_t) = \left[\Sigma(S_t) - \Sigma(\hat{S}_t) \right] + \left[(I - \mathcal{P})\Sigma(\hat{S}_t) \right].$$

This grouping is what forced the eigenvalue tail to be a property of $\hat{S}_t$. Regroup the same four terms differently — add and subtract $\mathcal{P}\Sigma(S_t)$ instead of $\mathcal{P}\Sigma(\hat{S}_t)$:

$$\Sigma(S_t) - \mathcal{P}\Sigma(\hat{S}_t) = \underbrace{(I - \mathcal{P})\Sigma(S_t)}_{\text{a function of the full process only}} + \underbrace{\mathcal{P}\left[\Sigma(S_t) - \Sigma(\hat{S}_t)\right]}_{\text{Lipschitz-controlled by } \|e_t\|}.$$

Check: $(I - \mathcal{P})\Sigma(S_t) + \mathcal{P}\Sigma(S_t) - \mathcal{P}\Sigma(\hat{S}_t) = \Sigma(S_t) - \mathcal{P}\Sigma(\hat{S}_t)$. ✓ Algebraically identical decomposition of the same quantity — nothing is lost, only re-grouped.

The payoff of this regrouping: the first term no longer refers to $\hat{S}_t$ at all. It is a property of the full model, computable (in principle, and in closed form for some models — see §7.6) *before* the reduced model is even simulated.

7.2 The averaging convention that closes Gap 1

Definition. Let

$$\bar{A} := \frac{1}{T} \int_0^T \mathbb{E} \left[\Sigma(S_t) \Sigma(S_t)^\top \right] dt \in \mathbb{R}^{d \times d}.$$

This is a single, fixed, symmetric positive semi-definite matrix — the time-and-path average of the full model's instantaneous covariance along its own (unreduced) law. It does not reference $\hat{S}$, $\mathcal{P}$, or any reduced quantity, so there is no circularity in defining it. Diagonalize it exactly as in Section 3:

$$\bar{A} = \sum_{j=1}^d \bar{\lambda}_j \bar{u}_j \bar{u}_j^\top, \quad \bar{\lambda}_1 \geq \dots \geq \bar{\lambda}_d \geq 0.$$

Define $\mathcal{P}$ to be the orthogonal projector onto $\text{span}\{\bar{u}_1, \dots, \bar{u}_{d'}\}$ — i.e., build the reduced model from the eigenstructure of $\bar{A}$, not from the instantaneous $A = \Sigma \Sigma^\top$ at any single point. This is the offline step of Algorithm KCV-MC, made precise.

7.3 Proposition (this choice of $\mathcal{P}$ is provably optimal, not arbitrary)

Claim. Among all rank- d' orthogonal projectors Q on $\mathbb{R}^d$,

$$\text{tr} \left((I - Q) \bar{A} \right) \geq \sum_{j>d'} \bar{\lambda}_j,$$

with equality if and only if $Q = \mathcal{P}$ (the top- d' eigenspace of $\bar{A}$).

Proof. Since $\text{tr}(\bar{A})$ is fixed, minimizing $\text{tr}((I - Q)\bar{A}) = \text{tr}(\bar{A}) - \text{tr}(Q\bar{A})$ over rank- d' orthogonal projectors Q is equivalent to maximizing $\text{tr}(Q\bar{A})$. This is exactly **Ky Fan's maximum principle** (a classical fact in matrix analysis, e.g. Bhatia, *Matrix Analysis*, Ch. III): for any symmetric matrix A with eigenvalues $\lambda_1 \geq \dots \geq \lambda_d$,

$$\max_{\substack{Q \text{ orthogonal projector} \\ \text{rank}(Q) = d'}} \text{tr}(Q\bar{A}) = \sum_{j=1}^{d'} \bar{\lambda}_j,$$

attained exactly when $\text{range}(Q)$ is the top- d' eigenspace of $\bar{A}$. Substituting,

$$\min_Q \text{tr}((I - Q)\bar{A}) = \text{tr}(\bar{A}) - \sum_{j=1}^{d'} \bar{\lambda}_j = \sum_{j>d'} \bar{\lambda}_j,$$

attained at $Q = \mathcal{P}$. ■

Why this matters: this is the same variational principle underlying PCA and POD. It says the “spectral criterion” branding is earned here, not decorative — $\mathcal{P}$ built from A 's top- d' eigenspace isn't just a candidate reduced basis, it's the one that literally minimizes the discarded diffusion energy that drives the whole bound. No other choice of rank- d' projector does better against this metric.

7.4 The regrouped Grönwall bound, derived in full

Start exactly where Section 5.1 starts, with $e_t = S_t - \hat{S}_t$ and $e_0 = \mathcal{P}S_0 - \hat{S}_0$; assume the natural initial condition $\hat{S}_0 = \mathcal{P}S_0$, so $e_0 = 0$ (consistent with the centered-setup convention already used in Section 4). Itô's formula on $\|e_t\|^2$ gives, exactly as before,

$$\frac{d}{dt} \mathbb{E} \|e_t\|^2 = 2 \mathbb{E} \left\langle e_t, b(S_t) - \mathcal{P}b(\hat{S}_t) \right\rangle + \mathbb{E} \|\Sigma(S_t) - \mathcal{P}\Sigma(\hat{S}_t)\|_{HS}^2.$$

Drift term. Split $b(S_t) - \mathcal{P}b(\hat{S}_t) = (I - \mathcal{P})b(S_t) + \mathcal{P} \left[b(S_t) - b(\hat{S}_t) \right]$ (identical regrouping trick, applied to the drift). Using Young's inequality $2xy \leq x^2 + y^2$ on the first piece and the Lipschitz bound $\|\mathcal{P} [b(S_t) - b(\hat{S}_t)]\| \leq L_b \|e_t\|$ on the second (using $\|\mathcal{P}x\| \leq \|x\|$ since $\mathcal{P}$ is an orthogonal projector):

$$2 \left\langle e_t, b(S_t) - \mathcal{P}b(\hat{S}_t) \right\rangle \leq \|e_t\|^2 + \|(I - \mathcal{P})b(S_t)\|^2 + 2L_b \|e_t\|^2 = (1 + 2L_b) \|e_t\|^2 + \|(I - \mathcal{P})b(S_t)\|^2.$$

Diffusion term. Split $\Sigma(S_t) - \mathcal{P}\Sigma(\hat{S}_t) = (I - \mathcal{P})\Sigma(S_t) + \mathcal{P}\left[\Sigma(S_t) - \Sigma(\hat{S}_t)\right]$ (the §7.1 regrouping). Using $\|a + b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$ and the Lipschitz bound $\|\Sigma(S_t) - \Sigma(\hat{S}_t)\|_{HS} \leq L_\Sigma \|e_t\|$:

$$\|\Sigma(S_t) - \mathcal{P}\Sigma(\hat{S}_t)\|_{HS}^2 \leq 2\|(I - \mathcal{P})\Sigma(S_t)\|_{HS}^2 + 2L_\Sigma^2 \|e_t\|^2.$$

Combine. Adding both pieces and taking expectations,

$$\frac{d}{dt} \mathbb{E} \|e_t\|^2 \leq \underbrace{(1 + 2L_b + 2L_\Sigma^2)}_{=:K'} \mathbb{E} \|e_t\|^2 + \underbrace{\mathbb{E} \|(I - \mathcal{P})b(S_t)\|^2 + 2\mathbb{E} \|(I - \mathcal{P})\Sigma(S_t)\|_{HS}^2}_{=:g(t)}.$$

Both terms of $g(t)$ now depend **only on the full process** S_t — the circularity is gone. By Grönwall's inequality with $e_0 = 0$,

$$\mathbb{E} \|e_T\|^2 \leq e^{KT} \int_0^T g(t) dt.$$

Evaluating the diffusion integral exactly. Since $\mathcal{P}$ is a fixed (deterministic, time-independent) matrix, it commutes with expectation and time-integration:

$$\int_0^T \mathbb{E} \|(I - \mathcal{P})\Sigma(S_t)\|_{HS}^2 dt = \text{tr} \left((I - \mathcal{P}) \int_0^T \mathbb{E} [\Sigma(S_t) \Sigma(S_t)^\top] dt \right) = \text{tr} \left((I - \mathcal{P}) T \bar{A} \right) = T \sum_{j>d'} \bar{\lambda}_j,$$

using $\bar{A}$'s definition and the same trace/eigenbasis computation as in the original Lemma 3 (§4.3) — this step is now an **exact identity**, not an approximation, precisely because $\mathcal{P}$ was built from A 's own eigenstructure (§7.3).

The drift integral does not reduce to a trace identity in the same way — $b(\cdot)$ is a vector field, not a covariance operator, so there's no natural “spectrum” for it. Define it as its own well-posed quantity,

$$\bar{m} := \frac{1}{T} \int_0^T \mathbb{E} \|(I - \mathcal{P})b(S_t)\|^2 dt,$$

the *discarded drift energy* — a legitimate, finite, a-priori-defined number, just not a spectral one.

7.5 Theorem (Gap-1-Resolved General Spectral Criterion)

Statement. Under the general Standing Assumptions of Section 7 (Lipschitz, bounded coefficients $b(\cdot)$, $\Sigma(\cdot)$ with Lipschitz constants L_b, L_Σ), Lipschitz payoff f , synchronous coupling, and initial condition $\hat{S}_0 = \mathcal{P}S_0$, with $\mathcal{P}$ the top- d' eigenspace of

$$\bar{A} = \frac{1}{T} \int_0^T \mathbb{E} \left[\Sigma(S_t) \Sigma(S_t)^\top \right] dt,$$

$$1 - \rho^2 \leq \frac{L_f^2 e^{-2rT}}{\sigma_H^2} \cdot e^{K'T} T \left(\bar{m} + 2 \sum_{j>d'} \bar{\lambda}_j \right), \quad K' = 1 + 2L_b + 2L_\Sigma^2.$$

Proof. Chain Lemma 1 (§4.1, unchanged — never used constancy), Lemma 2 (§4.2, unchanged — same reason), and the bound $\mathbb{E} \| S_T - \hat{S}_T \|^2 \leq e^{K'T} T \left(\bar{m} + 2 \sum_{j>d'} \bar{\lambda}_j \right)$ derived in §7.4, exactly as the three lemmas were chained in §4.4. ■

This is a genuine, fully general (state-dependent-coefficient) theorem. Gap 1, as originally posed — “which average defines the spectrum” — has one canonical, non-circular, provably-optimal answer.

7.6 What this does — and does not — achieve, stated with full honesty

1. **Gap 1 is closed.** There is no remaining ambiguity about which spectrum to use: it is A 's, defined via the full (unreduced) process, and that choice is provably optimal by Ky Fan's principle, not a convention picked for convenience.
2. **This does not recover the exact constant-coefficient identity as a special case — it is strictly looser even there.** Set $L_b = L_\Sigma = 0$ (truly constant coefficients). Then $K' = 1$, not 0 — a spurious constant survives from the Young's-inequality step, which was never needed in the original Lemma 3 (there, $b(S_t) - \mathcal{P}b(\hat{S}_t) = (I - \mathcal{P})b$ deterministically, no cross term at all). Similarly the factor of 2 on $\sum_{j>d'} \bar{\lambda}_j$ has no counterpart in the exact identity. **This is an honest cost of the elementary (Itô + Young + Grönwall) method used to sidestep Gap 2**

rather than solving it with semigroup perturbation theory as the original roadmap intended. A future pass using Trotter–Kato-type estimates could plausibly remove both the spurious constant and the factor of 2, but that is not done here — Gap 2 is sidestepped, not closed.

3. **Gap 3 (the e^{KT} blow-up) is untouched.** The bound above is still a finite-horizon guarantee only; nothing in this section addresses long-maturity behavior.
4. **Heston is still not covered — for a reason worth stating clearly.** Heston’s variance process is a CIR (square-root) diffusion: $dv_t = \kappa(\theta - v_t)dt + \xi\sqrt{v_t}dW_t$. The diffusion coefficient $\xi\sqrt{v_t}$ is **not globally Lipschitz** near $v_t = 0$ — this is a textbook fact about square-root diffusions, and it’s exactly the condition your own Section 2 table already excludes under “Local-volatility singularities: coefficients not globally Lipschitz/bounded.” So even this newly-derived, more general theorem does not rigorously apply to Heston as stated; it would need a localization argument (e.g. stopping at a boundary, or using known moment bounds for CIR processes in place of a global Lipschitz constant) before the standing assumptions are satisfied. **This means the caveat requested in §5.3 needs to be stated for both theorems, not just the original one.**
5. **Computability of $\bar{A}$ is model-dependent, not automatic.** For Heston specifically (setting aside point 4 for a moment), $\Sigma(S_t)\Sigma(S_t)^\top$ depends only on the instantaneous variance v_t , whose first moment solves a closed-form linear ODE ($\mathbb{E}[v_t] = \theta + (v_0 - \theta)e^{-\kappa t}$) — so $\bar{A}$ is, in that specific model, cheaply computable in closed form without any Monte Carlo, which is a genuinely attractive practical feature once the Lipschitz issue in point 4 is separately handled. For general local-vol or SABR-type models, no such closed form is likely to exist, and $\bar{A}$ would need to be estimated via a cheap pilot simulation or an asymptotic expansion — this is a legitimate remaining implementation question, not a mathematical gap.
6. **The drift term $\bar{m}$ breaks the “purely spectral” framing.** It’s a well-defined, finite, computable quantity, but it isn’t an eigenvalue tail — there’s no natural notion of “spectrum” for a vector field. Three honest options going forward: present $\bar{m}$ as an explicit secondary correction term (weakens the clean spectral-only headline, but is fully honest); restrict the theorem’s stated scope to driftless or risk-neutral-drift-only models where $\bar{m}$ vanishes or is controlled separately (narrows the claim but keeps the spectral framing clean); or fold b into an augmented generator/covariance-like object so a single spectral quantity captures both drift and diffusion

discard (mathematically the most elegant option, but not attempted here — a genuine next step, not a closed question).

7.7 Exact list of remaining points to investigate

This is the precise, ranked list of what's still open after this revision — not a restatement of the original three gaps, but what's actually left once Gap 1 is closed:

1. **Verify the Heston Lipschitz issue directly (point 4 above) before using Heston anywhere as a “covered” example.** Either add a localization/truncation argument (e.g. using the Feller condition $2\kappa\theta \geq \xi^2$ and known CIR moment bounds to replace the global Lipschitz assumption with a local one plus a boundary-crossing-probability estimate), or explicitly reclassify Heston as an *empirically tested, not theoretically covered* case throughout the paper — abstract, experiments section, and conclusion alike. → **Status: a full localization/truncation strategy is now worked out in Section 10.**
2. **Decide how to present the looseness identified in point 2 above (§7.6).** Either attempt the semigroup-perturbation refinement to tighten K' and remove the factor-of-2, or explicitly report the looser constants and, ideally, show numerically how much slack they introduce relative to the true (simulated) $1 - \rho^2$ on a test case where both the constant- and general-coefficient bounds can be computed side by side.
3. **Work out $\bar{A}$ in closed form for each model class you intend to test** — done above for Heston's first moment; still needed for any local-vol or SABR variant used in the numerical section, including a decision on Monte-Carlo-based estimation error if no closed form exists. → **Status: Heston and lognormal SABR done in full closed form in Section 11; a decision framework for the general case is given there too.**
4. **Resolve the drift term's role (point 6 above)** — pick one of the three options and carry it consistently through the abstract's headline claim, since “spectral criterion” as a title promise is only fully accurate if the drift term is either eliminated by scope or folded into a genuinely spectral object.
5. **Gap 3 (finite horizon) remains completely open** — no work in this document touches it; it is exactly as open as it was in the original draft.
6. **State the technical integrability conditions explicitly.** The interchange of expectation and time-differentiation in the Itô step, and of expectation and trace in §7.4, is justified here by the boundedness of b , Σ already assumed in Section 7's standing assumptions — but the write-up should say this in one explicit sentence rather than leaving it implicit, for full rigor. → **Status: done in full in Section 9.**

8. Where this sits relative to prior work

The paper’s own novelty audit (Section 8 of the draft) compares against reduced-basis / greedy-variance-minimization approaches (e.g. Cohen–Boyaval–Heider–Lelièvre 2010) and deterministic-volatility control variates (Du–Liu–Gu 2014; Fouque–Han). The stated convergent finding across the prior literature is that **none of it supplies an a priori bound on ρ computable before simulation** — existing methods all measure effectiveness empirically, after the fact, via output variance. If that comparison holds up under a full literature pass (worth double-checking directly against each paper rather than only the literature-search summaries), the “predictive, not reactive” framing is a genuinely defensible novelty claim — but only for the case actually proven (constant coefficients), which is the gap discussed in Section 5.1 of this document.

9. Point 6 — The Integrability and Regularity Conditions Behind Section 7, Made Explicit

Throughout Section 7, several standard technical steps were used without being spelled out: exchanging expectation and time-differentiation in Itô’s formula, exchanging expectation and time-integration in the trace computation, and the basic well-posedness of the SDEs involved. None of these are hard, given the Standing Assumptions already in place (Lipschitz, globally bounded b, Σ) — but “routine given the assumptions” is different from “stated.” Here is the complete list, each with its justification.

9.1 Well-posedness of $S, \hat{S}$, and the coupled pair

Claim. Under Lipschitz, globally bounded $b, \Sigma: \mathbb{R}^d \rightarrow \mathbb{R}^d, \mathbb{R}^{d \times d}$, both S and $\hat{S}$ have unique strong solutions on $[0, T]$, and the pair $(S, \hat{S})$, sharing the driving Brownian motion W , is well-defined on a common filtered probability space.

Justification. Global Lipschitz plus (trivially implied, since bounded $\Rightarrow$ linear growth) linear growth is exactly the hypothesis of the classical existence-and-uniqueness theorem for Itô SDEs (Øksendal, *Stochastic Differential Equations*, Thm. 5.2.1; Karatzas–Shreve, Thm.

5.2.9), giving a pathwise-unique strong solution S . For $\hat{S}$: since $\mathcal{P}$ is an orthogonal projector, $\|\mathcal{P}\|_{\text{op}} = 1$, so

$$\|\mathcal{P}b(x) - \mathcal{P}b(y)\| \leq \|b(x) - b(y)\| \leq L_b \|x - y\|,$$

and identically for $\Sigma - \mathcal{P}b, \mathcal{P}\Sigma$ inherit the same Lipschitz constants, so the same theorem applies to $\hat{S}$'s equation. Since $S, \hat{S}$ share the same W , the pair solves one $(d + d')$ -dimensional SDE whose stacked coefficients inherit Lipschitz continuity from the components — the same theorem gives well-posedness of the joint law, exactly what “synchronous coupling” presumes.

9.2 A priori second-moment bound

Claim. $\sup_{t \in [0, T]} \mathbb{E} \|S_t\|^2 < \infty$, and identically for $\hat{S}_t$.

Justification. Itô on $\|S_t\|^2$ gives

$$d\|S_t\|^2 = \left[2\langle S_t, b(S_t) \rangle + \|\Sigma(S_t)\|_{HS}^2 \right] dt + 2S_t^\top \Sigma(S_t) dW_t. \text{ Using the global bound}$$

$\|b\|_\infty, \|\Sigma\|_\infty \leq C$ and Young's inequality, $2\langle S_t, b(S_t) \rangle \leq C^2 + \|S_t\|^2$, so the drift of $\|S_t\|^2$ is bounded by $C'(1 + \|S_t\|^2)$. Applying the localization argument of §9.3 below (stop, bound, let the stopping level go to infinity via Fatou) gives $\mathbb{E} \|S_t\|^2 \leq e^{C't} (\|S_0\|^2 + C't) < \infty$ on $[0, T]$.

9.3 The localization argument behind §7.4's Grönwall step

This is the one place a genuine gap could hide, so it's worth doing in full.

The issue. Itô's formula gives $d\|e_t\|^2 = [\dots]dt + 2e_t^\top \sigma_t dW_t$. Passing to

$\frac{d}{dt}\mathbb{E} \|e_t\|^2 = \mathbb{E}[\dots]$ requires the stochastic integral $\int_0^t 2e_s^\top \sigma_s dW_s$ to be a *true* martingale,

needing $\mathbb{E} \int_0^T \|e_s^\top \sigma_s\|^2 ds < \infty$ — not obviously true before you already know $\mathbb{E} \|e_s\|^2$

is finite. A circularity.

The fix. Define $\tau_n := \inf\{t \geq 0: \|e_t\| \geq n\}$. On $[0, t \wedge \tau_n]$, $\|e_s\| \leq n$ by construction and σ_s is bounded (global bound $\|\Sigma\|_\infty$), so $\|e_s^\top \sigma_s\| \leq n \|\Sigma\|_\infty$ is bounded — the stochastic integral up to τ_n is a true martingale, no circularity. Applying Itô and taking

expectations on $[0, t \wedge \tau_n]$ rigorously reproduces §7.4's Grönwall bound for $\mathbb{E} \|e_{T \wedge \tau_n}\|^2$, with a constant K' that **does not depend on n** — the key feature that makes this base case easy, in contrast to Heston (§10), where the analogous local Lipschitz constant grows with n .

Removing the truncation. Since b, Σ are globally bounded, e_t cannot explode in finite time, so $\tau_n \rightarrow \infty$ a.s., and $e_{T \wedge \tau_n} \rightarrow e_T$ a.s. by path continuity. Fatou's lemma gives

$$\mathbb{E} \|e_T\|^2 = \mathbb{E} \left[\lim_n \|e_{T \wedge \tau_n}\|^2 \right] \leq \liminf_n \mathbb{E} \|e_{T \wedge \tau_n}\|^2 \leq e^{K'T} T \left(\bar{m} + 2 \sum_{j>d'} \bar{\lambda}_j \right),$$

exactly §7.5's Theorem, now with the martingale-exchange step justified rather than assumed.

9.4 Fubini for the trace identity, and continuity of the integrand

Both are easy given boundedness, but worth naming. (i) $t \mapsto \mathbb{E} [\Sigma(S_t) \Sigma(S_t)^\top]$ is bounded (by $\|\Sigma\|_\infty^2$) on the compact interval $[0, T]$; interchanging $\int_0^T \mathbb{E}[\cdot] dt$ with $\mathbb{E} \int_0^T (\cdot) dt$, and further with the fixed linear functional $\text{tr}(\cdot)$, are elementary applications of Fubini for finite-dimensional matrix entries over a bounded interval — no measure-theoretic subtlety. (ii) Continuity of $t \mapsto \mathbb{E} [(I - \mathcal{P})b(S_t)]$ follows from continuity of b (implied by Lipschitz), path continuity of S_ρ and dominated convergence using the global bound $\|b\|_\infty$.

9.5 Non-degeneracy

Claim. $\sigma_H^2 = \text{Var}(\Pi) > 0$ should be a stated standing assumption, not an implicit one.

Why. Lemma 1 (§4.1) divides by σ_H^2 ; if Π is a.s. constant, ρ is undefined and the whole criterion is vacuous. This fails only in genuinely degenerate cases (constant f , zero-volatility model), so it's a mild condition — but it belongs in the assumptions list, not left silent.

Summary of Point 6. With (9.1)–(9.5) made explicit, the derivation in §7.4–7.5 is fully rigorous *as literally stated* — under global boundedness and Lipschitz continuity. The next two sections show that almost no realistic volatility model actually satisfies those literal conditions, and what to do about it.

10. Point 1 — A Genuine Proof Strategy for Heston, via Localization and a Tail Bound

10.1 Where exactly Heston breaks the Standing Assumptions

Work in log-price coordinates $Z_t = (X_t, v_t)$, $X_t = \log S_t$ (justified here, and further in §11):

$$dX_t = (r - 1/2v_t) dt + \sqrt{v_t} dW_t^1, \quad dv_t = \kappa(\theta - v_t) dt + \xi\sqrt{v_t} dW_t^2, \quad d\langle W^1, W^2 \rangle_t = \rho dt.$$

The drift $b(Z) = (r - 1/2v, \kappa(\theta - v))$ is linear, hence globally Lipschitz — no issue. The diffusion involves $\sqrt{v}$, and $\frac{d}{dv}\sqrt{v} = \frac{1}{2\sqrt{v}} \rightarrow \infty$ as $v \rightarrow 0^+$, so $\Sigma(Z)$ is **not globally**

Lipschitz near $v = 0$ — exactly the condition the paper’s own exclusion table already names under “local-volatility singularities.” Existence and uniqueness of the CIR/Heston SDE is not itself in question — classical, via Yamada–Watanabe for Hölder-1/2 diffusion coefficients (Ikeda–Watanabe, *Stochastic Differential Equations and Diffusion Processes*, Ch. IV). What’s in question is whether §7.4’s specific proof technique — a global-Lipschitz Grönwall argument — can be adapted. It can, via localization, but not for free.

10.2 The strategy: localize on a safe annulus, then bound the escape probability

Fix $n \in \mathbb{N}$ and define

$$\tau_n := \inf \left\{ t \geq 0 : v_t \notin [1/n, n] \text{ or } \hat{v}_t \notin [1/n, n] \right\} \wedge T,$$

where $\hat{v}_t$ is the variance-coordinate of $\hat{S}_t$ under the natural embedding of $\text{range}(\mathcal{S}) \subset \mathbb{R}^2$.

Step 1 — a valid bound on the stopped process. On $[1/n, n]$, $v \mapsto \sqrt{v}$ has Lipschitz

constant $L_n = \frac{1}{2\sqrt{1/n}} = \frac{\sqrt{n}}{2}$ (maximal slope, at the left endpoint). So on $[0, \tau_n]$, §7.4’s

entire argument applies verbatim with L_Σ replaced by L_n :

$$\mathbb{E} \| e_{T \wedge \tau_n} \|^2 \leq e^{K_n T} T \left(\bar{m}_n + 2 \sum_{j > d'} \bar{\lambda}_j^{(n)} \right), \quad K_n' = 1 + 2L_b + 2L_n^2 = 1 + 2L_b + n/2,$$

with $\bar{m}_n, \bar{\lambda}_j^{-(n)}$ computed from the stopped (hence bounded, well-defined) process. K_n' grows only linearly in n — $e^{K_n T}$ grows exponentially in n , not worse, which matters for Step 3.

Step 2 — controlling the escape event. Write

$\mathbb{E} \|e_T\|^2 = \mathbb{E} \left[\|e_T\|^2 1_{\tau_n \geq T} \right] + \mathbb{E} \left[\|e_T\|^2 1_{\tau_n < T} \right]$. The first term is bounded by Step 1 (since $e_{T \wedge \tau_n} = e_T$ on $\{\tau_n \geq T\}$). For the second, Cauchy–Schwarz gives

$$\mathbb{E} \left[\|e_T\|^2 1_{\tau_n < T} \right] \leq \sqrt{\mathbb{E} \|e_T\|^4} \cdot \sqrt{\mathbb{P}(\tau_n < T)}.$$

Both factors are controllable using CIR/Heston-specific facts:

- $\mathbb{E} \|e_T\|^4 < \infty$: in log-price coordinates, X_T has finite moments of every order (the well-known Heston moment-explosion phenomenon afflicts the *price level* $S_T = e^{X_T}$, not X_T itself — see Andersen–Piterbarg, *Moment explosions in stochastic volatility models*, Finance & Stochastics 2007), and v_T has finite moments of every order via its non-central-chi-squared law (Cox–Ingersoll–Ross 1985). A fixed finite $M := \mathbb{E} \|e_T\|^4$ exists, independent of n .
- $\mathbb{P}(\tau_n < T) \rightarrow 0$: the Feller condition $2\kappa\theta \geq \xi^2$ gives $v_t > 0$ a.s. for every $t \in [0, T]$ simultaneously; on the compact interval, a continuous a.s.-positive path attains a positive minimum, so $\inf_{t \leq T} v_t > 0$ a.s. Downward continuity of probability then gives $\mathbb{P}(\inf_{t \leq T} v_t < 1/n) \rightarrow 0$. The upper tail $\mathbb{P}(\sup_{t \leq T} v_t > n) \rightarrow 0$ follows from Markov’s inequality applied to CIR’s known exponential moment $\mathbb{E}[e^{\alpha v_t}]$ (finite below a threshold set by the Riccati-ODE blow-up boundary of the CIR moment-generating function) via a maximal inequality on the associated supermartingale. A full Chernoff rate is extractable from the same machinery but isn’t needed for the qualitative conclusion below.

Step 3 — the bound exists, for every fixed n . Combining Steps 1–2,

$$\mathbb{E} \|e_T\|^2 \leq e^{K_n T} T \left(\bar{m}_n + 2 \sum_{j > d'} \bar{\lambda}_j^{-(n)} \right) + \sqrt{M} \cdot \sqrt{\mathbb{P}(\tau_n < T)},$$

a genuinely valid, finite bound for every fixed n — no limit needs to be taken. Chaining through Lemmas 1–2 exactly as in §7.5 turns this into a valid bound on $1 - \rho^2$ for Heston.

10.3 What this does and doesn't accomplish

- **It closes the existence question.** A valid, finite, rigorous bound on $1 - \rho^2$ exists for Heston, constructed explicitly — “Heston isn't covered” becomes “Heston is covered, via truncation.” Real progress, not a restatement of the gap.
 - **It doesn't give a clean constant.** The two terms in Step 3 trade off oppositely as n grows (Grönwall term grows like e^{cn} , escape term shrinks). Finding the optimal n^* is a genuine but mechanical follow-up — closer to an afternoon of algebra than an open question, but not done here.
 - **A sharper version of this already exists in the literature, under a different name.** This is structurally the same question studied in the strong-convergence-rate literature for CIR/Heston discretization schemes (Alfonsi, *High order discretization schemes for the CIR process*, Math. Comp. 2013; Dereich–Neuenkirch–Szpruch, *On strong approximation for the CIR process*, 2012; Cozma–Reisinger, *Strong convergence rates for Euler approximations to a class of stochastic path-dependent volatility models*, 2018). Those use a Yamada–Watanabe-type comparison argument (a sequence of smooth approximations to $|x|$, tailored to the square-root coefficient) that avoids the crude Cauchy–Schwarz split above and yields a genuinely optimized rate. Importing that machinery is the natural next step for a fully sharp Heston bound, and — since the target estimates already exist and would need adaptation rather than invention — it's a well-defined, bounded piece of work, not an open-ended one.
-

11. Point 4 — Worked Closed-Form Spectral Calculations

11.1 Heston, in full closed form

Using §10.1's log-price coordinates, the Cholesky decomposition $\Sigma(Z_t) = \sqrt{v_t} L$,

$L = \begin{pmatrix} 1 & 0 \\ \xi\rho & \xi\sqrt{1-\rho^2} \end{pmatrix}$, gives an exact factorization:

$$\Sigma(Z_t)\Sigma(Z_t)^\top = v_t C, \quad C := LL^\top = \begin{pmatrix} 1 & \xi\rho \\ \xi\rho & \xi^2 \end{pmatrix}.$$

The key simplification: the entire time-dependence of the Heston covariance is carried by the scalar CIR process v_t , multiplying one fixed matrix C . Since

$\mathbb{E}[v_t] = \theta + (v_0 - \theta)e^{-\kappa t}$ is closed form, so is its time average:

$$\bar{v} := \frac{1}{T} \int_0^T \mathbb{E}[v_t] dt = \theta + (v_0 - \theta) \frac{1 - e^{-\kappa T}}{\kappa T},$$

and therefore

$$\bar{A}_{\text{Heston}} = \bar{v} C = \bar{v} \begin{pmatrix} 1 & \xi\rho \\ \xi\rho & \xi^2 \end{pmatrix}.$$

Because $\bar{A}$ is a scalar times a fixed matrix, its eigenvectors are C 's eigenvectors, and its eigenvalues are $\bar{v}$ times C 's:

$$\bar{\lambda}_{1,2} = \bar{v} \cdot \frac{(1 + \xi^2) \pm \sqrt{(1 - \xi^2)^2 + 4\xi^2\rho^2}}{2}$$

(sanity check: at $\rho = 0$, C is diagonal and this reduces to $\bar{\lambda} \in \{\bar{v}, \bar{v}\xi^2\}$ exactly, as it must).

Heston's state is 2-dimensional, so a $d' = 1$ reduction discards exactly $\sum_{j>1} \bar{\lambda}_j = \bar{v} \cdot \lambda_-(C)$ —

fully explicit, computable with a 2×2 eigenvalue formula, no simulation.

Caveat tying back to Point 1. This computation of $\bar{A}$ is valid on its own terms — it only used finiteness of CIR moments, unrelated to the Lipschitz question. But *plugging it into the §7.5 Theorem* is only justified once §10's localization repair is carried through; this section tells you what the bound evaluates to, not by itself that it rigorously applies.

11.2 Lognormal SABR ($\beta = 1$): cleaner, but not assumption-free

In log-forward coordinates $Y_t = \log F_t$, the same Cholesky trick gives $\Sigma(Z_t)\Sigma(Z_t)^\top = \sigma_t^2 C'$, $C' = \begin{pmatrix} 1 & \nu\rho \\ \nu\rho & \nu^2 \end{pmatrix}$, and since σ_t is a driftless GBM, $\mathbb{E}[\sigma_t^2] = \sigma_0^2 e^{\nu^2 t}$ exactly, giving

$$\bar{A}_{\text{SABR}, \beta=1} = \sigma_0^2 \frac{e^{\nu^2 T} - 1}{\nu^2 T} C',$$

fully closed form, by the identical mechanism as Heston.

A genuine correction to how I framed this earlier. I initially suggested this model sidesteps the Lipschitz problem — it doesn't, quite. In log-coordinates the diffusion $\Sigma(Z) = \sigma L'$ is globally Lipschitz (linear in σ), better than Heston. But the Itô-correction drift term is $b_1(Y, \sigma) = -1/2\sigma^2$ — **quadratic in σ , not globally Lipschitz**, just for a different reason (unbounded growth, not a near-origin singularity). Switching to price-level coordinates doesn't remove the problem, it relocates it: drift becomes trivially zero, but the diffusion's $(1, 1)$ -entry becomes the bilinear term σF , not globally Lipschitz jointly in (F, σ) either.

The general pattern this reveals. No natural coordinate choice makes a genuine multiplicative-stochastic-volatility model satisfy global Lipschitz + bounded coefficients — the failure just relocates between drift and diffusion. So the localization technique built for Heston in §10 isn't a Heston-specific patch; it's the **generic repair this whole model class needs** (Heston, SABR at any β , essentially any local-vol or stochastic-vol model used in practice). SABR's version is easier to execute than Heston's — *polynomial growth at infinity* rather than a *near-boundary singularity*, needing only a one-sided ball-truncation plus a standard Lyapunov argument (e.g. $V(\sigma) = 1 + \sigma^2$, $\mathcal{L}V \leq CV$) rather than CIR boundary analysis — but the same three-step strategy (truncate, bound the stopped process, bound the escape probability) carries over directly.

Practical recommendation. Revise §7's Standing Assumptions from “globally bounded, Lipschitz” to “locally Lipschitz, at most polynomial growth, plus a Lyapunov/non-explosion condition” — the standard framework for this model class (Khasminskii-type conditions) — with the localization argument built **once**, as a named, reusable lemma. §10's Heston derivation and this section's SABR sketch then become two applications of one lemma, not two separate patches.

11.3 General local vol / SABR with $\beta \neq 1$: no closed form, a decision framework

For $\beta \notin \{0, 1\}$, SABR isn't affine, and cross-moments like $\mathbb{E}\left[F_t^{2\beta}\sigma_t^2\right]$ have no known closed form — exactly why practitioners use asymptotic expansions (Hagan et al., *Managing Smile Risk*, Wilmott 2002) rather than exact formulas. Three honest options, in decreasing rigor and increasing practicality:

1. **Small-vol-of-vol asymptotic expansion** of $\bar{A}$ to leading order in ν , using the same singular-perturbation machinery behind Hagan's formula — genuine additional work (redone for $\bar{A}$ specifically, not a citation away), with a characterizable error order.

2. **Pilot Monte Carlo estimate.** Simulate $N_{\text{pilot}} \ll N$ paths of the full model offline, estimate $\hat{A}$ empirically, and control the eigenvalue/eigenvector error via standard matrix concentration (matrix Bernstein $\Rightarrow$ Weyl's inequality for eigenvalues, Davis–Kahan $\sin\theta$ for eigenvectors) — off-the-shelf machinery giving an honest “ $\mathcal{P}$ is within ϵ of optimal with probability $1 - \delta$ ” statement.
3. **Restrict scope for the first paper.** State plainly that closed-form $\bar{A}$ is available for the affine subclass (Heston, $\beta = 1$ SABR, and more generally any model where the state-dependent part of $\Sigma \Sigma^\top$ factors as (scalar affine process) $\times$ (fixed matrix) — a genuine, nameable class, not just two examples), with the general case flagged as documented future work. The most honest and least effortful option, consistent with the paper's existing practice of labeling scope boundaries clearly.

11.4 The Heston moment cascade, completed

§11.1 only needed $\mathbb{E}[v_t]$ for $\bar{A}$ itself. Two things elsewhere still need higher CIR moments: the discarded-drift energy $\bar{m}$ (§7.4, §12.2), and the fourth-moment integrability ingredient in the Heston localization argument (§10.2). Both are covered by the same mechanism, carried further here than before.

The general recursion. For $dv_t = \kappa(\theta - v_t)dt + \xi\sqrt{v_t}dW_t$ and $m_n(t) := \mathbb{E}[v_t^n]$, Itô's formula on v_t^n gives $dv_t^n = [n\kappa\theta v_t^{n-1} - n\kappa v_t^n + 1/2n(n-1)\xi^2 v_t^{n-1}]dt + n\xi v_t^{n-1/2}dW_t$, so taking expectations,

$$\dot{m}_n = -n\kappa m_n + [n\kappa\theta + 1/2n(n-1)\xi^2]m_{n-1}.$$

This is a **linear cascade**: each m_n solves a first-order linear ODE driven only by the *already-solved* m_{n-1} , so every moment is obtainable in closed form, in sequence, via an integrating factor.

$n = 1$: $\dot{m}_1 = -\kappa m_1 + \kappa\theta \Rightarrow m_1(t) = \theta + (v_0 - \theta)e^{-\kappa t}$ (as in §11.1, and §9's earlier derivation).

$n = 2$: $\dot{m}_2 = -2\kappa m_2 + (2\kappa\theta + \xi^2)m_1$. Integrating factor $e^{2\kappa t}$; the particular solution driven by m_1 's two terms is

$$m_2^{\text{part}}(t) = \frac{(2\kappa\theta + \xi^2)\theta}{2\kappa} + \frac{(2\kappa\theta + \xi^2)(v_0 - \theta)}{\kappa}e^{-\kappa t},$$

plus a homogeneous term $Ce^{-2\kappa t}$ fixed by $m_2(0) = v_0^2$. **Sanity check:** as $t \rightarrow \infty$,
 $m_2 \rightarrow \theta^2 + \frac{\xi^2 \theta}{2\kappa}$ — matching the known CIR stationary second moment exactly
 $(\text{Var}_\infty(v) = \xi^2 \theta / 2\kappa, \text{ so } m_{2,\infty} = \text{Var}_\infty + \theta^2)$. ✓.

$n = 3, 4$: identical mechanism. $\dot{m}_3 = -3\kappa m_3 + (3\kappa\theta + 3\xi^2)m_2$ and
 $\dot{m}_4 = -4\kappa m_4 + (4\kappa\theta + 6\xi^2)m_3$, each a linear ODE driven by the previous, each solvable
via the integrating factor $e^{n\kappa t}$ into a combination of $\{1, e^{-\kappa t}, \dots, e^{-n\kappa t}\}$ with coefficients
fixed by matching $m_n(0) = v_0^n$ against the particular-solution recursion. No new technique
is needed past $n = 2$ — only bookkeeping.

What this feeds. All the time-averages needed elsewhere are now closed form via

$$\int_0^T e^{-j\kappa t} dt = \frac{1 - e^{-j\kappa T}}{j\kappa}:$$

$$\bar{v} = \frac{1}{T} \int_0^T m_1 dt = \theta + (v_0 - \theta) \frac{1 - e^{-\kappa T}}{\kappa T} \quad (\S 11.1\text{'s } \bar{v}, \text{ confirmed}),$$

and for an affine discarded-drift piece $(a + c v_t)$ (§12.2's Heston computation),

$$\int_0^T \mathbb{E} \left[(a + c v_t)^2 \right] dt = \int_0^T (a^2 + 2ac m_1 + c^2 m_2) dt \text{ — fully closed form using only } m_1, m_2$$

above. The SABR analogue is simpler still, since α_t is lognormal: $\mathbb{E}[\alpha_t^n] = \alpha_0^n e^{n(n-1)\nu^2 t/2}$,
closed form for every n directly.

11.5 Drift-term handling, precisely scoped

§12.2 already showed $\bar{m}$ vanishes exactly for pure risk-neutral-tradable-asset drift
 $(b(x) = rx)$. Restated with the mechanism made explicit: since $\mathcal{P}b(x) = \mathcal{P}(rx) = r\mathcal{P}x$, the
reduced model inherits $d\hat{S}_t = r\hat{S}_t dt + \mathcal{P}\Sigma dW_t$ exactly, so $(I - \mathcal{P})b(S_t) = r(I - \mathcal{P})S_t$ is a
residual that the Grönwall argument absorbs into the exponential rather than into a
separate $\bar{m}$ term — **no discarded-drift energy is left over.**

This does **not** extend to Heston or SABR: their auxiliary-factor drift $(\kappa(\theta - v))$, or SABR's
driftless-but-log-quadratic structure) is not proportional to the state, so $\bar{m}$ is genuinely
nonzero there — but, per §11.4, fully computable in closed form via the moment cascade,
using only m_1 and m_2 .

12. Evaluating a Proposed Refinement — What Holds Up, What Needed Correcting, and One New Genuine Result

A follow-up proposal addressed three points: resolving Heston via localization, eliminating the drift term $\bar{m}$ via the risk-neutral measure, and closing Gap 2 via semigroup perturbation. Each is checked on its own merits below, the same way I'd want a collaborator to check mine — two needed a real correction, not a cosmetic one, and one, once actually derived rather than asserted, turns out to be a genuine and useful addition.

12.1 The Heston localization argument: the mechanics are right, the conclusion overclaims

The proposed stopping-time / annulus-truncation strategy is structurally identical to §10.2 — same τ_R , same bounded/tail split, same Cauchy–Schwarz step. That part is correct and simply restates what §10 already established.

The conclusion doesn't follow, though. The proposal states that by choosing R sufficiently large, the tail error becomes vanishingly small and the criterion holds to arbitrary precision. This conflates two different things: the *tail term* shrinking (true) and the *total bound* shrinking (false) — because the **bounded-part term also depends on R , and it grows**, not shrinks, as R increases.

Concretely: $K_R' = 1 + 2L_b + R/2$ grows linearly in R (§10.2), so $e^{K_R'T}$ grows exponentially in R ; and $\bar{m}_R, \sum \bar{\lambda}_j^{(R)}$, computed on the stopped process, themselves grow at least linearly in R (variance on the annulus $[1/R, R]$ scales with R). So the bounded-part term behaves like $A e^{cRT}$ for fixed $A, c > 0$. The tail term behaves like $B e^{-\alpha R/2}$ for a fixed $\alpha > 0$, set by CIR's moment-generating-function domain — not something that grows alongside R . The total,

$$\text{Total}(R) \sim A e^{cRT} + B e^{-\alpha R/2},$$

diverges to $+\infty$ as $R \rightarrow \infty$ — the growing term eventually dominates the shrinking one. “Choose R large” does the opposite of what the proposal claims.

The correct statement (already reached more carefully in §10.3): for *every fixed R* , the two-term sum is a valid, finite bound — that part is real and useful. The right move is to *minimize* $\text{Total}(R)$ over finite R (a standard, if not closed-form, one-dimensional optimization), not to send $R \rightarrow \infty$. The resulting bound is finite and computable, but **not** “arbitrarily small ϵ ” — it's some fixed, generally non-tiny number reflecting the real cost of

an elementary technique. A bound that genuinely vanishes, or is meaningfully tightened, needs the sharper CIR-specific comparison technique already cited in §10.3 (Alfonsi 2013; Dereich–Neuenkirch–Szpruch 2012; Cozma–Reisinger 2018) — truncation alone doesn't get there no matter how large R is chosen.

Verdict: mechanics correct, conclusion incorrect — corrected here, not incorporated as originally stated.

12.2 Eliminating the drift term: correct for a real special case, once properly derived

The algebraic observation is right and useful: if the full model's drift is exactly $b(S) = rS$ (every coordinate of S is itself a risk-neutral tradable asset — no separate latent factor), then since $\mathcal{P}$ is linear,

$$(I - \mathcal{P})b(S_t) = (I - \mathcal{P})(rS_t) = r(I - \mathcal{P})S_t,$$

correctly, and the reduced model inherits the same drift $(\mathcal{P}b(\hat{S}_t) = r\hat{S}_t$, since

$\hat{S}_t \in \text{range}(\mathcal{P})$). But the proposal stops at the algebra and asserts $\bar{m}$ is “absorbed into the spectral tail” without deriving how. Here's the derivation, done properly — and it comes out **better** than claimed, in this scoped case.

Proposition (exact drift handling when $b(S) = rS$). With $S_0 = \hat{S}_0 \in \text{range}(\mathcal{P})$, $e_t = S_t - \hat{S}_t$ satisfies

$$de_t = r e_t dt + \left[\Sigma(S_t) - \mathcal{P}\Sigma(\hat{S}_t) \right] dW_t$$

exactly — no discarded-drift residual at all, since $b(S_t) - \mathcal{P}b(\hat{S}_t) = r(S_t - \hat{S}_t) = r e_t$ precisely. Applying Itô and the same diffusion-side regrouping as §7.4 (unaffected by the drift — that's a separate, still-present difficulty):

$$\frac{d}{dt} \mathbb{E} \| e_t \|^2 \leq (2r + 2L_\Sigma^2) \mathbb{E} \| e_t \|^2 + 2h(t), \quad h(t) := \mathbb{E} \| (I - \mathcal{P})\Sigma(S_t) \|_{HS}^2,$$

and Grönwall gives

$$\mathbb{E} \| e_T \|^2 \leq e^{(2r+2L_\Sigma^2)T} \cdot 2T \sum_{j>d'} \bar{\lambda}_j.$$

A genuine improvement, not a restatement: the constant improves from the general case's $K' = 1 + 2L_b + 2L_\Sigma^2$ to $2r + 2L_\Sigma^2$ — the spurious “+1” from the drift's Young's-inequality split disappears (no cross-term is left to bound), and L_b is replaced by the risk-free rate r , typically small. And $\bar{m}$ doesn't get “absorbed” — it is **exactly zero**, since the discarded-drift residual vanishes identically, not approximately.

But this holds only for the case the proposal didn't state explicitly: pure multi-asset (basket-style) local-volatility models, where every coordinate of S is a tradable asset with drift $rS_t^{(i)}$, and there is no latent factor. It does **not** apply to Heston or SABR, whose variance/vol-of-vol coordinate has risk-neutral drift $\kappa(\theta - v_t)$ — genuinely not of the form $r \times$ (that coordinate). Concretely, for Heston (§11.1's coordinates $Z = (X, v)$, $d' = 1$, discarded eigenvector u_2 of C):

$$(I - \mathcal{P})b(Z_t) = u_2[u_2^\top a + u_2^\top BZ_t], \quad a = \begin{pmatrix} r \\ \kappa\theta \end{pmatrix}, \quad B = \begin{pmatrix} 0 & -1/2 \\ 0 & -\kappa \end{pmatrix},$$

a generically nonzero affine function of Z_t — no algebraic reason for its second moment to vanish, and it doesn't. It's still computable in closed form, via the same moment-cascade technique used for $\bar{v}$ in §11.1: $\mathbb{E}[v_t], \mathbb{E}[v_t^2], \mathbb{E}[X_t], \mathbb{E}[X_t v_t], \mathbb{E}[X_t^2]$ each solve a linear ODE, in sequence, driven by the previously-solved ones.

Worked piece: $\mathbb{E}[v_t^2]$ for CIR, closed form. Itô on v_t^2 :

$$dv_t^2 = [2\kappa\theta v_t - 2\kappa v_t^2 + \xi^2 v_t]dt + 2\xi v_t^{3/2}dW_t. \text{ With } p(t) := \mathbb{E}[v_t^2] \text{ and } m(t) = \theta + (v_0 - \theta)e^{-\kappa t} \text{ (known),}$$

$$p'(t) + 2\kappa p(t) = (2\kappa\theta + \xi^2)m(t),$$

a linear ODE, solved via integrating factor $e^{2\kappa t}$:

$$p(t) = v_0^2 e^{-2\kappa t} + (2\kappa\theta + \xi^2) \left[\frac{\theta}{2\kappa} (1 - e^{-2\kappa t}) + \frac{v_0 - \theta}{\kappa} (e^{-\kappa t} - e^{-2\kappa t}) \right].$$

The remaining pieces ($\mathbb{E}[X_t], \mathbb{E}[X_t v_t], \mathbb{E}[X_t^2]$) follow the identical mechanism — Itô, take expectation, linear ODE in terms of already-solved lower moments, integrating factor — mechanical from here, not conceptually new.

Verdict: correct and valuable once derived — but scoped to pure-tradable-asset (basket/local-vol) models. For Heston/SABR, $\bar{m}$ remains real and nonzero; it's computable in closed form via the cascade above, but doesn't vanish, and still needs one of §11.2's three honest options.

12.3 Gap 2 via semigroup perturbation: the right direction, not yet a proof

The instinct here isn't new — §5.1 already names semigroup perturbation theory (Trotter–Kato / Hille–Yosida) as the required machinery for Gap 2. What's submitted points at the right neighborhood but isn't a derivation, and calling it “solved theoretically” overstates what's there. Four specific gaps:

1. **Domain mismatch.** $\mathcal{L}$ acts on functions on $\mathbb{R}^d$; the reduced generator naturally acts on functions on $\text{range}(\mathcal{P})$. For $\mathcal{L} - \tilde{\mathcal{L}}$ to be well-defined, $\tilde{\mathcal{L}}$ must first be extended to the same space as $\mathcal{L}$ (e.g. via precomposition with $\mathcal{P}$) — unspecified, and the choice affects everything downstream.
2. **Regularity for Duhamel.** The variation-of-constants formula needs $\tilde{u}(s, \cdot)$ in the domain of $\mathcal{L}$ (or a core) for each s , and both semigroups strongly continuous on a common space — establishing this *is* the substance of a Trotter–Kato argument, not a preliminary to skip.
3. **The “no cross-terms” claim is asserted, not shown — and a direct check suggests it's false as stated.** Expanding $(\mathcal{L} - \tilde{\mathcal{L}})g(x)$ under the natural extension gives a difference of the form $\Sigma(x)\Sigma(x)^\top - \mathcal{P}\Sigma(\mathcal{P}x)\Sigma(\mathcal{P}x)^\top\mathcal{P}^\top$ — comparing the covariance *at* x against the covariance *at the projected point* $\mathcal{P}x$. This mixes exactly the two pieces §7.1's regrouping already separated (a discarded-mode piece and a Lipschitz-difference piece); it doesn't obviously isolate only the discarded modes. A fully worked expansion could still turn out favorable, but it isn't shown, and a first-pass check doesn't support the claim as stated.
4. **The translation step is missing.** Even granting an operator-level bound, connecting it back to the specific pathwise quantity $\mathbb{E} \|S_T - \hat{S}_T\|^2$ that Lemmas 2–3 are built around needs additional work not present in the sketch.

Verdict: the right tool is correctly named — consistent with §5.1's own roadmap — but this specific sketch has real, identifiable gaps and at least one likely-false claim (point 3). Gap 2 remains open, now with a precise list (points 1–4) of what a rigorous version needs to nail down first.

13. Closing Gap 3 — An Exact Result for Mean-Reverting (Ornstein–Uhlenbeck / Vasicek-Type) Models

Gap 3 has been open, untouched, through every revision so far: the general bound's $e^{K'T}$ factor blows up as the horizon T grows, so nothing proven so far says anything useful about long-dated instruments. This section closes it — not for the fully general nonlinear case, but for a realistic, named model class where an *exact* (not Grönwall-approximated) argument is available: multi-factor **mean-reverting Gaussian models** — Vasicek/Hull–White-type multi-curve rate models, mean-reverting basis or spread trades — exactly the “multi-curve interest-rate products” the paper’s own introduction (§1) names as a motivating high-dimensional problem.

13.1 Setup

$$dS_t = -\Theta(S_t - \theta) dt + \Sigma dW_t,$$

with $\Theta \in \mathbb{R}^{d \times d}$ **symmetric positive definite** (genuine mean reversion in every direction, eigenvalues $\gamma_1, \dots, \gamma_d > 0$), $\theta \in \mathbb{R}^d$ a fixed long-run mean, and Σ **constant** (Gaussian/Vasicek-type — no state-dependence in the diffusion). Note this is *not* covered by Section 4 (which required b constant, not linear) nor cleanly by Section 7 (only via the lossy Grönwall route) — it’s a genuine third case, and it turns out to admit an exact solution because OU processes do.

Following §7’s convention, the reduced model evaluates the same drift at the projected point: $d\hat{S}_t = -\mathcal{P}\Theta(\hat{S}_t - \theta) dt + \mathcal{P}\Sigma dW_t$, with $\mathcal{P}$ still the top- d' eigenspace of $A = \Sigma\Sigma^\top$ (now genuinely a fixed matrix, no time-averaging needed — Ky Fan’s optimality argument from §7.3 still applies unchanged). Writing $\hat{S}_t = \mathcal{P}\hat{S}_t$, this simplifies to $d\hat{S}_t = -\hat{\Theta}\hat{S}_t dt + \mathcal{P}\Theta\theta dt + \mathcal{P}\Sigma dW_t$ where $\hat{\Theta} := \mathcal{P}\Theta\mathcal{P}$ — symmetric (since $\Theta, \mathcal{P}$ both are), and positive definite when restricted to $\text{range}(\mathcal{P})$ (for $x \in \text{range}(\mathcal{P}) \setminus \{0\}$: $x^\top \mathcal{P}\Theta\mathcal{P}x = x^\top \Theta x > 0$).

13.2 Exact solutions and the exact error identity

Both S_t and $\hat{S}_t$ are autonomous linear (OU) SDEs — critically, **each evolves independently of the other’s state**, coupled only through sharing the same W . Each has the standard closed-form OU solution (verified by checking $d\left[e^{\Theta t}(S_t - \theta)\right] = e^{\Theta t}\Sigma dW_t$, using that Θ commutes with $e^{\Theta t}$):

$$S_t = \theta + e^{-\theta t}(s_0 - \theta) + \int_0^t e^{-\theta(t-u)} \Sigma dW_u, \quad \hat{S}_t = \hat{\theta} + e^{-\hat{\theta} t}(\hat{S}_0 - \hat{\theta}) + \int_0^t e^{-\hat{\theta}(t-u)} \mathcal{P} \Sigma dW_u,$$

where $\hat{\theta}$ solves $\hat{\theta} \hat{\theta} = \mathcal{P} \theta \theta$ on $\text{range}(\mathcal{P})$. Subtracting, the deterministic terms collect into a computable function $D(t)$ (which vanishes under the same centered initial conditions as §4.3's Remark — $s_0 \in \text{range}(\mathcal{P}), \theta \in \text{range}(\mathcal{P})$), and the stochastic parts combine into a single Itô integral:

$$e_T = D(T) + \int_0^T \left[e^{-\theta(T-u)} \Sigma - e^{-\hat{\theta}(T-u)} \mathcal{P} \Sigma \right] dW_u.$$

By the Itô isometry (for matrix-valued deterministic integrands,

$$\mathbb{E} \left\| \int_0^T M(u) dW_u \right\|^2 = \int_0^T \|M(u)\|_{HS}^2 du, \text{ and substituting } v = T - u:$$

$$\mathbb{E} \|e_T\|^2 = \|D(T)\|^2 + \int_0^T \|e^{-\theta v} \Sigma - e^{-\hat{\theta} v} \mathcal{P} \Sigma\|_{HS}^2 dv.$$

This is exact — no Grönwall, no Lipschitz constant, no factor of 2. In the centered setup ($D \equiv 0$), it's a direct, clean generalization of §4.3's Lemma 3 identity to linear (not just constant) drift.

13.3 Gap 3 closes for this class

Since θ is symmetric positive definite, $\|e^{-\theta v}\|_{\text{op}} = e^{-\gamma_{\min} v} \rightarrow 0$ exponentially as $v \rightarrow \infty$ (its eigenvalues are exactly $e^{-\gamma_i v}$). Since $\hat{\theta}$ restricted to $\text{range}(\mathcal{P})$ is also positive definite, and $\mathcal{P} \Sigma$'s image lies entirely in $\text{range}(\mathcal{P})$, $e^{-\hat{\theta} v} \mathcal{P} \Sigma \rightarrow 0$ exponentially too. So the integrand decays exponentially in v , and

$$\lim_{T \rightarrow \infty} \mathbb{E} \|e_T\|^2 = \int_0^\infty \|e^{-\theta v} \Sigma - e^{-\hat{\theta} v} \mathcal{P} \Sigma\|_{HS}^2 dv =: \mathcal{E}_\infty < \infty.$$

The pathwise error converges to a finite stationary value instead of blowing up.

Chaining through Lemmas 1-2 (unchanged, as always), and noting the discount factor

$e^{-2rT} \rightarrow 0$ as well (for $r > 0$), the full bound $\frac{L_f^2 e^{-2rT}}{\sigma_\Pi^2} \mathbb{E} \|e_T\|^2 \rightarrow 0$ as $T \rightarrow \infty$ — a much

stronger conclusion than merely “doesn't explode.” **Gap 3 is closed for this model class,** not worked around.

13.4 The clean case: when drift and diffusion share an eigenbasis

If $\Theta = \sum_j \gamma_j u_j u_j^\top$ and $\Sigma = \sum_j \sigma_j u_j u_j^\top$ (with $\sigma_j^2 = \lambda_j$) share the *same* eigenbasis as A — a natural special case — everything decouples mode by mode. On retained modes ($j \leq d'$), $\hat{\Theta}$ and $\mathcal{P}$ act identically to Θ and the identity, so the two terms cancel exactly. On discarded modes ($j > d'$), $\mathcal{P}$ kills the reduced side entirely, leaving only $e^{-\Theta v}$'s contribution. This gives, for **any finite** T :

$$\mathbb{E} \|e_T\|^2 = \sum_{j>d'} \frac{\lambda_j}{2\gamma_j} (1 - e^{-2\gamma_j T}), \quad \mathcal{E}_\infty = \sum_{j>d'} \frac{\lambda_j}{2\gamma_j}.$$

This is a genuinely interpretable, still-spectral formula: **the long-run error is the discarded eigenvalue tail, weighted by the inverse of each mode's own mean-reversion rate**. A discarded mode that mean-reverts quickly (γ_j large) contributes little, even though it's never tracked — it corrects itself. A discarded mode that mean-reverts slowly contributes more. This is exactly the intuition a practitioner would want confirmed, and now it's proven, not assumed.

13.5 Consistency check: this is a genuine generalization of Section 4

As $\gamma_j \rightarrow 0^+$ for every discarded mode (no mean reversion at all — approaching Section 4's setting), L'Hôpital gives $\frac{1 - e^{-2\gamma_j T}}{2\gamma_j} \rightarrow T$, so $\mathbb{E} \|e_T\|^2 \rightarrow T \sum_{j>d'} \lambda_j$ — **exactly** Lemma 3's

original boxed identity (§4.3). Section 4's constant-drift result is recovered as the zero-mean-reversion limit of this new theorem. This is a strong internal consistency check, not a coincidence: it confirms the new derivation is a genuine, correctly-scoped extension rather than an unrelated result that happens to look similar.

13.6 Honest scope

This closes Gap 3 **only** for constant-diffusion, linear-mean-reverting-drift models — real and useful (multi-factor Vasicek/Hull-White short-rate models, mean-reverting spread or basis trades), but not the paper's headline stochastic-volatility targets. It does **not** apply to Heston, SABR, or local vol, where Σ is state-dependent and the exact Gaussian machinery above has no foothold. That said, Heston's own variance drift $\kappa(\theta - v_t)$ is mean-reverting — the same qualitative mechanism (fast-reverting discarded modes self-correct) plausibly extends there too, which would explain why Heston is used for long-dated options in

practice without the kind of blow-up Gap 3 warns about in the worst case. Making that rigorous for genuinely state-dependent diffusion would need a hypocoercivity-type argument (Villani’s framework for systems that are dissipative in some directions but merely driven, not dissipative, in others — exactly Heston’s cascade structure between v and X) — a well-posed, citable next step, not attempted here.

14. A Modest Tightening of Gap 2’s Constants for the General (Nonlinear) Case

For models outside Section 13’s reach — genuinely nonlinear drift or state-dependent diffusion, where exact Gaussian solvability isn’t available — the crude weighted split used throughout §7.4 (always splitting evenly, coefficient “(1, 1)” in the $\|a + b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$ sense) can be tightened with no new machinery, just a sharper elementary inequality.

Weighted version. For any $\delta, \epsilon > 0$, Young’s inequality ($2xy \leq \delta x^2 + y^2/\delta$) and its two-term generalization ($\|a + b\|^2 \leq (1 + \epsilon)\|a\|^2 + (1 + 1/\epsilon)\|b\|^2$) applied to the same drift/diffusion regroupings as §7.4 give

$$\mathbb{E} \|e_T\|^2 \leq e^{K'(\delta, \epsilon)T} T \left[\frac{1}{\delta} \bar{m} + (1 + \epsilon) \sum_{j>d'} \bar{\lambda}_j \right], \quad K'(\delta, \epsilon) = \delta + 2L_b + \left(1 + \frac{1}{\epsilon}\right) L_\Sigma^2.$$

Setting $\delta = \epsilon = 1$ recovers exactly §7.5’s original bound ($K' = 1 + 2L_b + 2L_\Sigma^2$) — this is a strict generalization, not a different result. Two things make it useful:

- **It recovers §4.3’s exact identity as a limit.** In the constant-coefficient case, $L_\Sigma = 0$ exactly (Σ is literally constant, so $\Sigma(S_t) - \Sigma(\hat{S}_t) \equiv 0$), and taking $\epsilon \rightarrow 0^+$ gives coefficient $(1 + \epsilon) \rightarrow 1$ with *no* penalty on K' (since $(1 + 1/\epsilon)L_\Sigma^2 = 0$ regardless of ϵ) — recovering the exact constant-coefficient result with a coefficient of exactly 1, not 2. This confirms the weighted family strictly contains the sharp answer as a special/limiting case, unlike the fixed-(1, 1) version.
- **It’s tunable per-problem.** For fixed ϵ , minimizing over δ is a one-variable calculus problem reducing to a quadratic in δ (via $\frac{d}{d\delta} \left[e^{\delta T} (\bar{m}/\delta + c) \right] = 0 \Rightarrow Tc\delta^2 + T\bar{m}\delta - \bar{m} = 0$), giving an explicit, if not

elegant, closed-form optimum; ϵ can then be tuned similarly or jointly by numerical search for a given $(T, L_b, L_\Sigma, \bar{m}, \text{spectral tail})$.

What this does not do. I checked whether pushing δ, ϵ far enough could make $K'(\delta, \epsilon) < 0$ — genuine long-horizon stability *without* the exact-solvability machinery of Section 13, via dissipativity of b alone. It can't, in general: even for the natural test case of a symmetric, positive-definite linear drift ($b(x) = -\Theta(x - \theta)$, exactly Section 13's setting), the AM-GM-optimal threshold works out to requiring $\gamma_{\min}(\Theta) > \sqrt{2} \gamma_{\max}(\Theta)$ — impossible unless Θ is a multiple of the identity, since $\gamma_{\min} \leq \gamma_{\max}$ always. The elementary weighted-Young's technique is too lossy to capture stability even in the one case (Section 13) where stability is genuinely true and provable — which is exactly why Section 13 needed the exact Itô-isometry argument instead of a sharper Grönwall. This is worth stating plainly rather than glossing over: **this section tightens constants; it does not, and structurally cannot, close Gap 3 on its own.**

15. A Lower Bound and Two-Sided Criterion (Constant-Coefficient Case)

Every bound so far (§4's Main Theorem, §13's OU extension) only upper-bounds $1 - \rho^2$: it guarantees “at least this much” reduction when the discarded tail is small. None of it says anything about the *failure* direction — if the tail is large, is the control variate provably useless, or might it work anyway? Answering that needs a genuine lower bound, not just a small upper bound.

15.1 The identity that makes this possible: $\hat{S}_T = PS_T$, exactly

Recall from §4.3: $S_T = s_0 + bT + \Sigma W_T$ and $\hat{S}_T = \mathcal{P}s_0 + \mathcal{P}bT + \mathcal{P}\Sigma W_T$. Applying $\mathcal{P}$ termwise to the first line reproduces the second, exactly:

$$\mathcal{P}S_T = \mathcal{P}s_0 + \mathcal{P}bT + \mathcal{P}\Sigma W_T = \hat{S}_T.$$

This is the fact that makes a clean lower bound possible. In the constant-coefficient regime, the reduced model's terminal state isn't approximately the projected true terminal state — it *is* the projected true terminal state, with no gap at all. (This identity is special to constant coefficients; in the state-dependent case $\hat{S}_T \neq \mathcal{P}S_T$ in general, which is a restatement of Gap 2 — see §15.5.)

15.2 Lemma 4 (conditional-variance ceiling)

Statement. For any random variable Y measurable with respect to a sub- σ -algebra $\mathcal{G}$,

$$\text{Corr}(\Pi, Y)^2 \leq \frac{\text{Var}\left(\mathbb{E}[\Pi \mid \mathcal{G}]\right)}{\text{Var}(\Pi)} =: R_{\max}^2(\mathcal{G}),$$

with equality iff Y is (a.s.) an affine function of $\mathbb{E}[\Pi \mid \mathcal{G}]$.

Proof. Since Y is $\mathcal{G}$ -measurable, the tower property gives

$$\mathbb{E}[\Pi Y] = \mathbb{E}\left[\mathbb{E}[\Pi Y \mid \mathcal{G}]\right] = \mathbb{E}\left[Y \mathbb{E}[\Pi \mid \mathcal{G}]\right], \text{ and similarly}$$

$\text{Cov}(\Pi, Y) = \text{Cov}\left(\mathbb{E}[\Pi \mid \mathcal{G}], Y\right)$. By the ordinary Cauchy–Schwarz inequality applied to this covariance,

$$\text{Cov}(\Pi, Y)^2 = \text{Cov}(\mathbb{E}[\Pi \mid \mathcal{G}], Y)^2 \leq \text{Var}\left(\mathbb{E}[\Pi \mid \mathcal{G}]\right) \cdot \text{Var}(Y),$$

with equality iff Y is an affine function of $\mathbb{E}[\Pi \mid \mathcal{G}]$ (the standard Cauchy–Schwarz equality condition). Dividing by $\text{Var}(\Pi)\text{Var}(Y)$ gives the stated bound on $\text{Corr}(\Pi, Y)^2$. ■

Reading it: among *all* possible payoffs built from the information in $\mathcal{G}$, the best possible correlation with Π is achieved by $\mathbb{E}[\Pi \mid \mathcal{G}]$ itself — the L^2 -projection of Π onto $\mathcal{G}$ -measurable random variables. Nothing built from $\mathcal{G}$ alone can beat that, no matter how cleverly constructed.

15.3 The two-sided criterion

Apply Lemma 4 with $\mathcal{G} = \sigma(\mathcal{P}S_T)$. By §15.1, $\hat{S}_T = \mathcal{P}S_T$ exactly, so $Y = e^{-rT}f(\hat{S}_T)$ is $\sigma(\mathcal{P}S_T)$ -measurable — genuinely, not approximately — and Lemma 4 applies directly to the actual Y the algorithm uses, no gap to bridge:

$$\rho^2 \leq R_{\max}^2\left(\sigma(\mathcal{P}S_T)\right).$$

By the law of total variance, $\text{Var}(\Pi) = \text{Var}\left(\mathbb{E}[\Pi \mid \mathcal{P}S_T]\right) + \mathbb{E}\left[\text{Var}(\Pi \mid \mathcal{P}S_T)\right]$, so $1 - R_{\max}^2 = \mathbb{E}\left[\text{Var}(\Pi \mid \mathcal{P}S_T)\right] / \text{Var}(\Pi)$. Combined with §4’s Main Theorem:

$$\frac{\mathbb{E}[\text{Var}(\Pi \mid \mathcal{P}S_T)]}{\text{Var}(\Pi)} \leq 1 - \rho^2 \leq \frac{L_f^2 e^{-2rT} T}{\sigma_{\Pi}^2} \sum_{j>d'} \lambda_j.$$

This holds for **any** payoff f with $\text{Var}(\Pi) < \infty$ — Lipschitz isn't even needed for the lower bound (only for the upper bound, via Lemma 2). It's a genuine two-sided sandwich: the upper bound (already proven) guarantees success when the discarded tail is small; the lower bound now guarantees failure when the discarded directions carry payoff-relevant variance the reduction can't see, regardless of how small the discarded *eigenvalue* tail happens to be.

15.4 Why this catches failures the eigenvalue-tail bound misses

The upper bound in §4 depends only on $\sum_{j>d'} \lambda_j$ — the *variance* discarded, independent of whether the payoff is *sensitive* to that variance. A payoff that happens to ignore the discarded directions entirely can have $\rho = 1$ even with a large discarded tail — the naive guess “ $1 - \rho^2 \geq c \sum_{j>d'} \lambda_j$ ” is false in general for exactly this reason. §15.3's lower bound is

different in kind: it depends on $\text{Var}(\Pi \mid \mathcal{P}S_T)$ — how much of Π 's *own* variance is left unexplained after conditioning on what the reduction retains — so it automatically accounts for payoff sensitivity rather than only spectral energy. A worst-of or rainbow-style payoff, whose value depends sharply on pairwise differences between assets, has large conditional variance given any low-dimensional $\mathcal{P}S_T$ precisely because those differences live in directions a low-rank reduction can't resolve — this is the mechanism the lower bound is built to catch, and §16 below sharpens exactly this point.

15.5 Honest scope

This is exact and rigorous **only where §15.1's identity holds — constant coefficients.**

For state-dependent coefficients, $\hat{S}_T \neq \mathcal{P}S_T$ in general (the two differ by exactly the commutator error that Gap 2 is about), so $Y = f(\hat{S}_T)$ is not $\sigma(\mathcal{P}S_T)$ -measurable and Lemma 4 doesn't apply to the actual estimator without an additional step: bounding $\delta_T := \mathcal{P}S_T - \hat{S}_T$ (note $\delta_T = \mathcal{P}e_T$, so $\|\delta_T\| \leq \|e_T\|$ — a strictly smaller quantity than the Section 7 error) and using it to relate ρ to $R_{\max}(\sigma(\mathcal{P}S_T))$ via a perturbation argument. I attempted to verify a specific version of this correction and could not fully confirm the constant it produces — this general-case extension of the two-sided criterion is flagged

here as a well-posed, promising, but **not yet verified** piece of further work, not claimed as closed.

16. Active-Subspace Refinement: When Is Plain PCA Suboptimal?

16.1 The question §7’s Ky Fan argument doesn’t answer

§7.3 proved that the top- d' eigenspace of $\bar{A}$ is the *provably optimal* choice for minimizing discarded **diffusion energy**, $\text{tr}\left((I - \mathcal{P})\bar{A}\right)$. But the paper’s actual objective is maximizing ρ (correlation), not minimizing discarded variance — are these the same thing? They are not, and the gap between them is worth making precise.

16.2 A clean example where PCA is provably suboptimal

Take a linear payoff $\Pi = e^{-rT} w^\top S_T$ and a rank-1 reduction $Y = e^{-rT} c^\top S_T$ (any direction c , not necessarily A ’s top eigenvector), in the constant-coefficient/Gaussian setting. Since $\text{Cov}(S_T) \propto A = \Sigma \Sigma^\top$,

$$\rho^2 = \frac{\text{Cov}(\Pi, Y)^2}{\text{Var}(\Pi)\text{Var}(Y)} = \frac{(w^\top A c)^2}{(w^\top A w)(c^\top A c)}.$$

This is exactly a Cauchy–Schwarz ratio in the A -weighted inner product $\langle x, y \rangle_A := x^\top A y$: $\rho^2 \leq 1$ always, with **equality iff** $c \propto w$ — i.e., a rank-1 reduction that simply *points at the payoff direction* achieves $\rho = 1$ exactly, regardless of how much spectral energy sits elsewhere in A . Plain PCA (built from A ’s own top eigenvector, ignoring w entirely) generically does *not* achieve this — it’s optimal for a payoff-agnostic objective (discarded energy) but not for the actual, payoff-aware objective (correlation).

16.3 Reconciling this with §7 rather than contradicting it

§7’s Ky Fan result is still correct as stated — it answers a genuinely different question (minimize discarded energy, with no payoff in view) and the Main Theorem’s *upper bound*

$$1 - \rho^2 \leq \frac{L_f^2 e^{-2rT} T}{\sigma_\Pi^2} \sum_{j>d'} \lambda_j$$

remains valid regardless of which $\mathcal{P}$ is used, since Lemma 2’s

Lipschitz bound $L_f = \|w\|$ is a payoff-agnostic worst case. §16.2 sharpens that worst-

case bound into a payoff-aware exact value when the payoff direction is known — the general lesson is that $1 - \rho^2$ is really governed by the **misalignment** between the retained subspace and the payoff's own sensitivity direction, not by discarded energy alone; discarded energy is what you're stuck bounding when you *don't* know the payoff direction in advance.

16.4 Generalizing to nonlinear payoffs: active subspaces

For general (not necessarily linear) f , replace the fixed direction w with the time-averaged expected gradient $\bar{g} := \frac{1}{T} \int_0^T \mathbb{E} \left[\nabla f(S_T) \mid \mathcal{F}_t \right] dt$, and take $\mathcal{P}$ to be the top- d' eigenspace of the payoff-weighted operator $\mathbb{E} \left[\nabla f \nabla f^\top \right]$ (weighted appropriately by A) rather than of A alone. This is precisely Constantine's **active subspace** construction (Constantine, *Active Subspaces*, SIAM 2015), applied here to the specific problem of predicting control-variate correlation rather than general dimension reduction — a real, useful connection to name, though the general nonlinear case needs the gradient-weighting made precise rather than simply asserted.

16.5 A d^* prescription

With payoff-weighted eigenvalues $a_j := \lambda_j \left(\bar{g}^\top u_j \right)^2$, the predicted shortfall from retaining only the top d' directions is $\approx \left(\sum_{j>d'} a_j \right) / \left(\sum_j a_j \right)$, so a target-tolerance reduction dimension can be *prescribed* rather than merely described:

$$d^* = \min \left\{ d' : \frac{\sum_{j>d'} a_j}{\sum_j a_j} \leq \tau_{\text{target}} \right\}.$$

Larger d' lowers the residual misalignment (and raises ρ) at the cost of a bigger anchor computation, subject to the $d' \leq 3$ ceiling the rational-Krylov step needs for machine-precision evaluation (§4's Algorithm KCV-MC, step 2).

16.6 Scope

This is clean and exact for **terminal (European-style) payoffs**, where w or $\bar{g}$ is a fixed direction. For path-dependent payoffs (Asian, barrier, autocallable), the relevant sensitivity direction genuinely varies along the path, so the optimal reduction subspace is time-

varying and a single fixed $\mathcal{P}$ is only an approximation — a real scope limit, not swept under the rug.

17. Summary verdict

	Assessment
Mathematical correctness of Section 4 (constant-coefficient Main Theorem)	Solid. Every step checks out; the trace identity in Lemma 3 is exact, not approximate.
Novelty of the <i>proven</i> content in isolation	Modest — the constant-coefficient case reduces to standard PCA + a direct Gaussian trace computation.
Two-sided criterion (lower bound)	New, and rigorous, for constant coefficients (Section 15): $\hat{S}_T = \mathcal{P}S_T$ exactly in this regime, which is what makes the conditional-variance lower bound apply to the <i>actual</i> estimator, not an oracle proxy. Closes the “one-directional only” caveat that’s been open since the first draft. General state-dependent case: attempted, not verified — flagged honestly rather than claimed.
Active-subspace refinement	New and correct (Section 16): plain PCA (§7) is optimal for discarded <i>energy</i>, provably <i>not</i> optimal for correlation when the payoff direction is known — a clean Cauchy–Schwarz counterexample, with a $d^\star$ prescription rule. Exact for terminal payoffs; approximate for path-dependent ones (stated as a real scope limit).
Gap 1 (averaging convention)	Closed (Section 7). The spectrum is defined via the time-and-path-averaged <u>covariance</u> of the <i>full</i> process, $\bar{A}$, and $\mathcal{P}$’s optimality is proven via Ky Fan’s maximum principle — not an arbitrary convention.

Assessment

Gap 2 (projection-dynamics commutation)

Closed exactly, but only for the Section 13 model class (constant Σ , linear mean-reverting drift) — the Itô-isometry identity there has no approximation at all, no factor of 2, no spurious constant. **Still open for genuinely nonlinear/state-dependent-diffusion models** (Heston, SABR, local vol); Section 14 gives a tunable but structurally limited tightening for that case, and Section 12.3's semigroup-perturbation critique still stands as the honest list of what a full resolution needs.

Gap 3 (finite-horizon blow-up)

Closed exactly for the Section 13 model class: the error provably converges to a finite stationary value as $T \rightarrow \infty$ (not just “doesn’t blow up” — genuinely vanishes once discounting is included), with a clean closed form $\sum_{j>d'} \lambda_j / (2\gamma_j)$ in the commuting case, and a verified consistency check recovering Section 4 exactly as mean-reversion $\rightarrow 0$. **Still open for Heston/SABR/local vol** — plausible by analogy (Heston’s own drift is mean-reverting) but not proven; would need a hypocoercivity-type argument, named but not executed (Section 13.6).

Assessment

Drift term $\bar{m}$

Resolved for a real special case: exactly zero when every state coordinate is a risk-neutral tradable asset with no latent factor (Section 11.5/12.2, properly derived, not merely asserted). **Nonzero for Heston/SABR, but now fully computable:** the CIR moment cascade (Section 11.4) supplies every moment needed in closed form, not just the first.

Heston coverage

A valid, finite, non-vanishing bound exists via localization for any fixed truncation level (Section 10), and the optimal-over-truncation-level bound is finite and computable — but it does **not** shrink to zero or “arbitrary precision” (Section 12.1 corrects this specific overclaim). Heston is “covered” in the sense of a genuine finite guarantee, not in the sense of a sharp or elegant one.

Closed-form spectral calculations

Done in full for Heston and lognormal SABR (Section 11), including the complete moment cascade (§11.4) — both reduce to (scalar affine process) $\times$ (fixed matrix), making A elementary. General $\beta \neq 1$ SABR / local vol has no closed form; a three-option decision framework is given (Section 11.3).

Assessment

Novelty of the *positioning* (a priori vs. reactive)

Strong, if the prior-art comparison holds up — this remains the paper’s real contribution. Now genuinely **two-sided** (Section 15) for the constant-coefficient case, which strengthens this claim materially.

Consistency risk to fix before submission

State explicitly, wherever Heston numerics are presented, that (a) they test the Section 5 conjecture, not a theorem, and (b) even the best available bound for Heston (Section 10–12.1) is a finite, non-tight guarantee, not a vanishing-error one. Also state, wherever Gap 2/3 closure or the two-sided criterion is mentioned, that these apply to constant-coefficient / Section 13’s Gaussian mean-reverting class specifically, not to the paper’s stochastic-volatility headline examples.

Bottom line: Gap 1 is closed; Gap 2 and Gap 3 are closed exactly for a real, named model class (Section 13); the criterion is now genuinely **two-sided** for constant coefficients (Section 15), with a correct, sharpened active-subspace refinement (Section 16) on top. That’s substantial, verified progress — but every one of these closures is scoped to constant-coefficient or linear-mean-reverting models, not the paper’s stochastic-volatility headline examples, and the paper should say so precisely everywhere these results are cited. The recurring lesson across every revision holds again here: check whether a “closed” claim is exact or approximate, and for which model class, before it goes in the abstract — and this round surfaced two more instances of that pattern (an inconsistent exponent in one localization argument, and a lower-bound claim that’s rigorous in the constant-coefficient case but was stated as if it extended further than verified).